MATS 12 · NEEL NANDA STREAM · EXPLORATION PHASE · 30 OCTOBER 2026

When do activation readers reveal more than the prompt?

Reporting gaps, causal read-site effects, and the limits of natural answer tracking.

Working hypothesis—not an established mechanism: a readout can be text-explainable and activation-sensitive without faithfully stating the subject's own changing state.

Guiv Farmanfarmaian · Suraj Prasad



decodable

a linear probe can read it

causal here

the answer depends on it at this token and layer

stated

the reader's text says it

Our 3 main measurements

OUR CLAIMS & CREDENCES

What we believe, and what could change our minds

Two observations are substantially firmer than our proposed explanations.

95% OBSERVATION **Structural state can survive yet go unreported.**

E100/E113; S22/S25 · Challenge: human-validated naming and independent decoding.

90% OBSERVATION **At the tested read token, causal effects rise late.**

E126 and unsteered E137 · Challenge: earlier-site effects on new tasks; registered natural rule failed.

60% HYPOTHESIS **Many successful readouts are explainable from prompt context.**

S06, E140; E107b/E116 pull against · Challenge: verified model-private tracking beyond matched text-only.

55% HYPOTHESIS **Training and output conventions help explain the silence.**

S29/S30; E139 remains mixed · Challenge: reward-only ablation or interface-matched SFT/RL study.

Provisional subjective credences—not statistical confidence intervals. Guiv and Suraj should agree on these numbers before submission.

DECISIVE STANDARD · A reader tracks a causally verified subject-private state better than a matched text-only predictor.

Sources: G-E100/E113/E126/E137/E140/E107b/E116; S-06/S22/S25/S29/S30; see claim ledger and appendix.

BEFORE THE SPRINT

What existed and what we kept

Exploration

- Readers replicated: NLA FVE 0.761; WorkspaceBench 12 of 12 inside our CIs
- The J-space survival test can't separate in-J from non-J features
- Only col (characters since newline) is cleanly non-J; no reader reads it
- Confirmatory run pre-registered, never run

Sprint

- Kept: the replication, best-of-11-layers, centred pushes, z against random directions
- Kept: WorkspaceBench measures re-solving the prompt (next slide)
- 23 Oct: Neel asked what one method finds that another can't
- That question, per concept, became the sprint

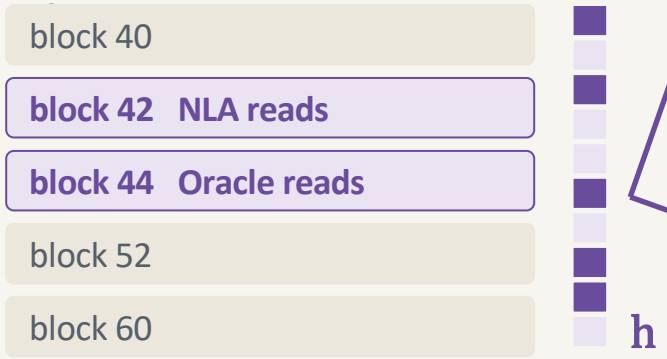
decodable · causal here · stated

An activation reader turns one residual vector into text

1 Prompt into Qwen3.6-27B

```
Archived question: The capital
of Peru is
End of archived question.
Current question: The capital
of France is
```

2 Residual output of one block, one



`block L output = hidden_states[L+1]`

3 Inject h into a reader that shares Qwen's weights

Oracle Lens · reads block 44

replaces a marker token's input embedding

$$e[\text{marker}] \leftarrow 16000 \cdot h / \|h\|$$

NLA · reads block 42

adds after its block 1, RL-trained so a critic can rebuild h

$$h_1[\text{marker}] \leftarrow h_1 + \|h_1\| \cdot h / \|h\|$$

4 Text

short continuations of a prompt

a paragraph explaining the activation

Read site Chat banks read the assistant-onset token, the last prompt token before Qwen would generate. Transfer chains (E133, E137) have no chat template and read the raw final " by". Never the first generated token.

Readers: Oracle Lens and NLA adapters on the same Qwen3.6-27B backbone · core/readers.py and read-site conventions in guivff/activation-readers @ c60fa970 (appendix A3)

decodable · causal here · stated

Three questions about every state: decodable, causal here, stated

Decodable

Can a linear probe read it from the activation?

held-out probe

AUROC

0.5 chance · 1.0 perfect

judge-free

Causal here

Does Qwen's answer depend on it at this token and layer?

patch the activation, watch the answer's logit difference

R

0 no effect · 1 full effect

judge-free

Stated

Does the reader's text say it?

blind judge; a YES must quote the readout

k / n

missing readouts reported

judged

Rule A state can be decodable without being causal here, and causal without being stated. These are never drawn on a shared axis; each later slide names its unit.

Judge: blind LLM judge, a YES must quote the readout · human audited later

decodable causal here stated

Readouts often look like a prompt of this kind, rebuilt and answered

Prompt to Qwen	Oracle Lens readout	NLA readout
What is the opposite of 'hot'? (Answer in Spanish, one word.)	- **Frío** - Frío - Frío (This is the single most common Spanish word for "cold" and the most direct translation)	Q&A format pattern: structured question-and-answer pair in English, now delivering a **Spanish-language direct answer**. … requiring the single Spanish word (e.g., "Frío") as the completion.
lang-02A	*Both state the answer, and the NLA names the answer language.*	
Eva carries the rope and Jonas carries the lantern. Who carries the lantern? Name only.	- **Alice** - **John** *(Note: The answer is strictly the name of the person who sleeps, as requested by …	Q&A format with strict answer-only instruction signals the model must provide just the name **"Alex"** as the direct answer to "Who sits first." …
own-02A	*Neither names Jonas or Eva. Both rebuild a prompt of this kind (a name-only question) and answer that.*	

Selected examples Two readouts picked to show the two styles, not rates; the rates are on the following slides. Excerpts shortened with …; highlights mark the words a judge would credit.

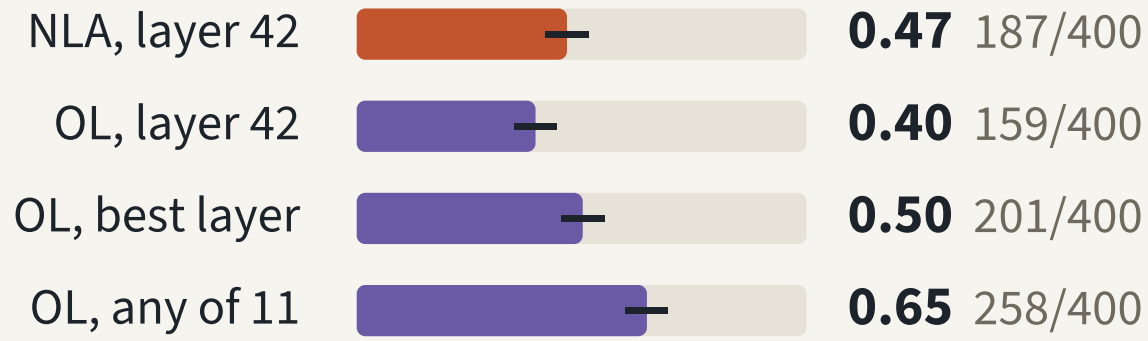
GATES · CHECKED BEFORE ANY READER RESULT

Check the readers first → they work as published

Reconstruction from text (FVE, 64 activations)



WorkspaceBench, regex pass (400 items, layer 42)



Gate 1 · the NLA works
Reproduces its published reconstruction: 0.761 vs the card's 0.766.

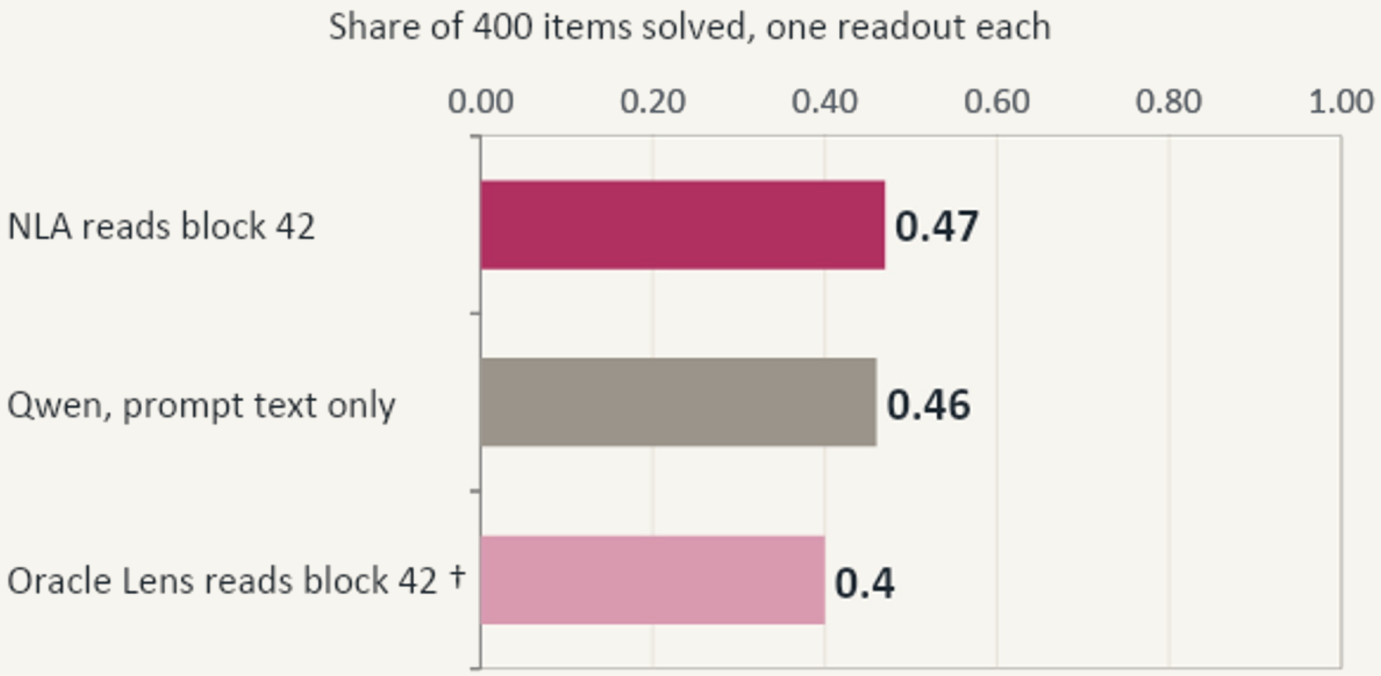
Gate 2 · Oracle Lens's lead
Published score: an item passes if any of 11 layers does. At one matched layer the NLA leads.

Potential · Method: the readers reproduce, so later gaps are about reading, not broken tools.

Wilson 95%. Any-of-11 adds +0.08 to +0.22 over the best single layer and is partly a best-of-11-samples bonus.
FVE: 95% bootstrap over 64 activations; the critic favours the NLA's own register.

decodable causal here stated

On WorkspaceBench, the NLA scores what a text-only copy of Qwen scores



95% CIs NLA 0.42–0.52 · text-only 0.41–0.51 · Oracle 0.35–0.45 † off the Oracle's trained grid (0.37 / 0.39 at 40 / 44)

NLA vs text-only: McNemar p = 0.75 (80 vs 75 discordant items)

NOT COMPUTE-MATCHED

Best-of-N scores, for reference

0.65 Oracle, published protocol: any of 11 layers (20–60) passes

0.69 text-only Qwen, best of 3 samples

0.96 any of 8 text-only samples (2 models × 2 templates)

0 of 187 NLA hits that every text-only sample missed

Does not show That the reader ignores the activation (at this layer the activation encodes the prompt), or equivalence: p = 0.75 is a failure to find a difference.

Suraj exp 06, 400 items, four multi-token families, item-bootstrap 95% CIs, McNemar on 80 vs 75 discordant items

decodable causal here stated

When Qwen's own answer changes with the wording, the reader rarely follows it

The test · template shown, realised pairs are private

```
What is the capital city of {entity}?
```

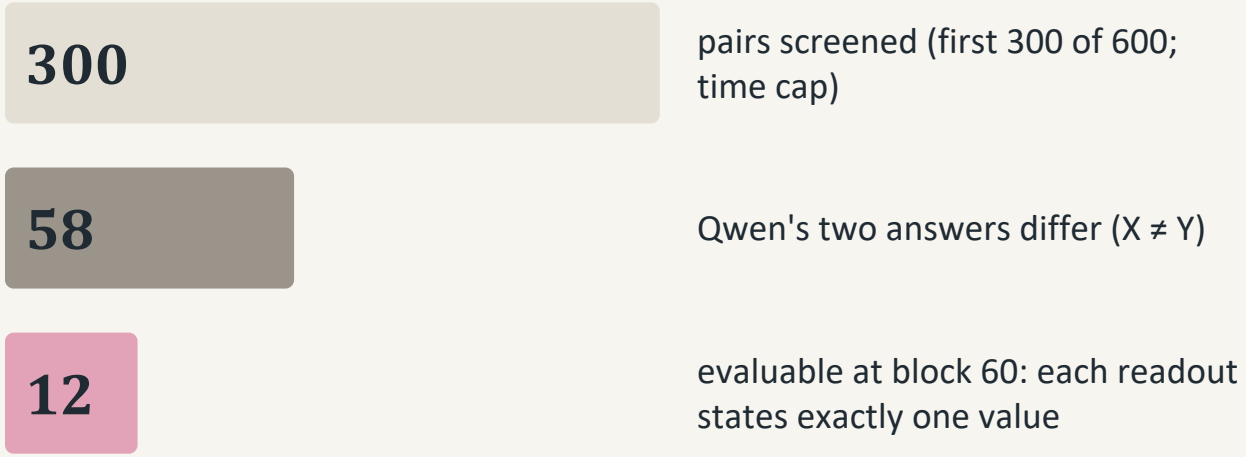
→ **Qwen answers X**

```
Which city is {entity}'s capital?
```

→ **Qwen answers Y (≠ X, by selection)**

Read both at block 60. Tracking: reader says X on wording A and Y on wording B. Same answer: one value for both.

Eligibility funnel · reader at block 60



Of 12 questions: 1 follows Qwen, 0.08 · 11 give one answer for both, 0.92

Does not show That the reader re-solves in general: rates are conditional on 12 of 58 pairs (7/116 readouts missing, 15 state several values). Paraphrases also move text-only models; a small natural difference may be below what the reader registers.

decodable causal here stated

After a whole-activation swap, content transfers readily; structural reporting is sparse

The intervention · Oracle Lens, block 44

recipient prompt

donor prompt

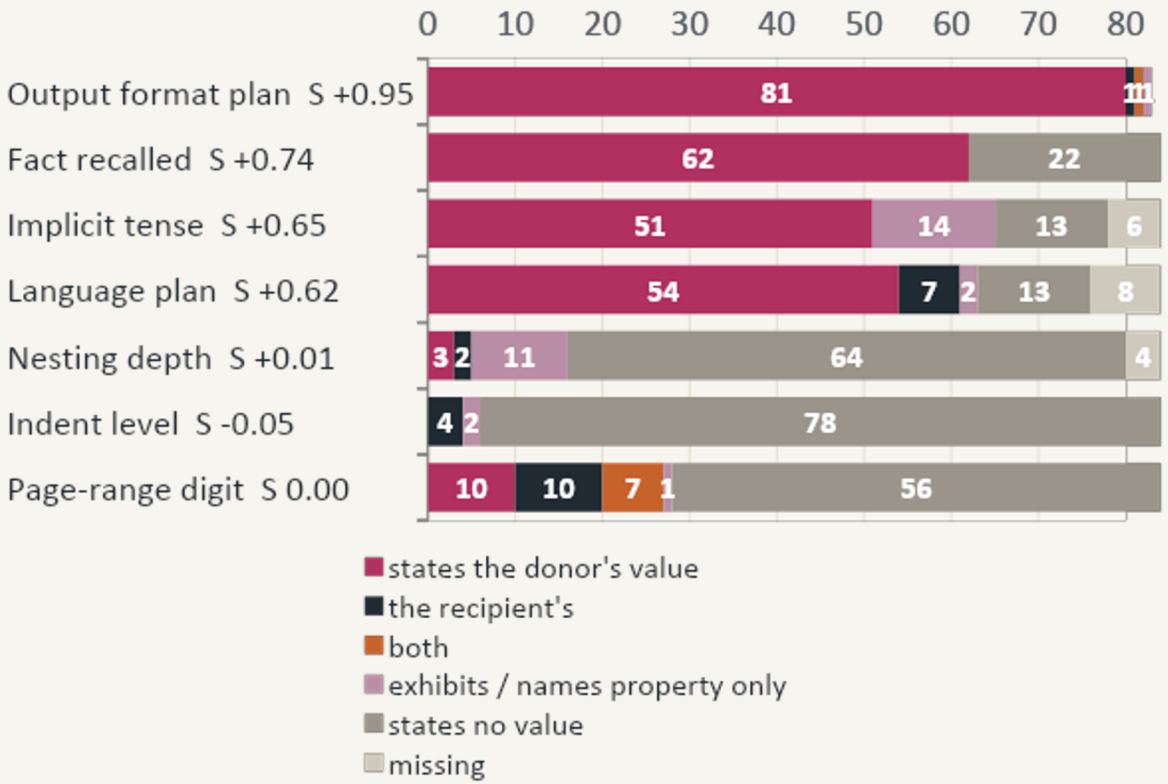
$h_{44}(\text{recipient}) \leftarrow h_{44}(\text{donor})$

→ reader writes text → judge: whose value does it state?

$$S = (\#\text{donor-only} - \#\text{recipient-only}) \,/\, \#\text{kept}$$

Content rows follow the donor. On nesting and indent, S ≈ 0 because the readout states no value at all (64 and 78 of 84). On the page digit it is partly cancellation: 10 donor, 10 recipient, 7 both.

Oracle readouts after the swap · counts of 84 per row

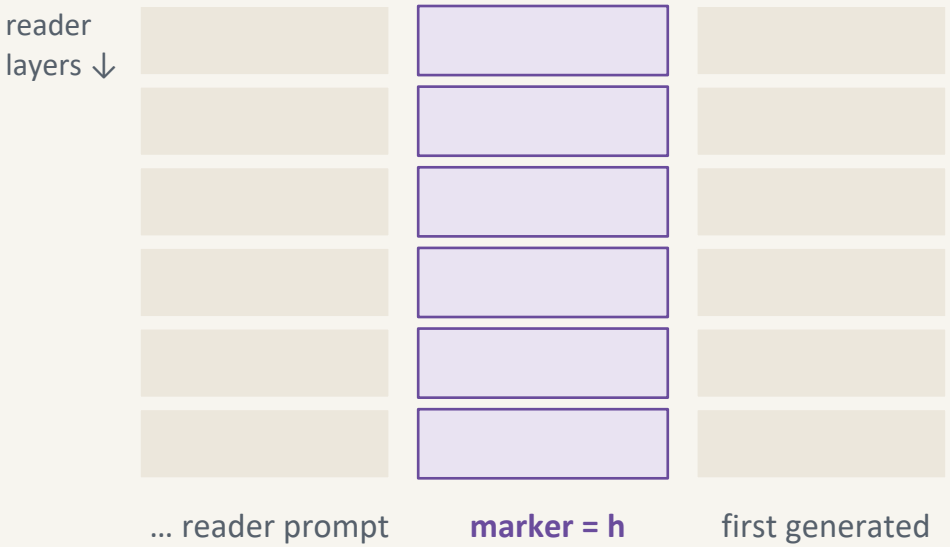


Does not show That the reader cannot see structure: probes read these rows at AUROC 1.00 (next slide). A transplant moves everything at once; the category census is post hoc.

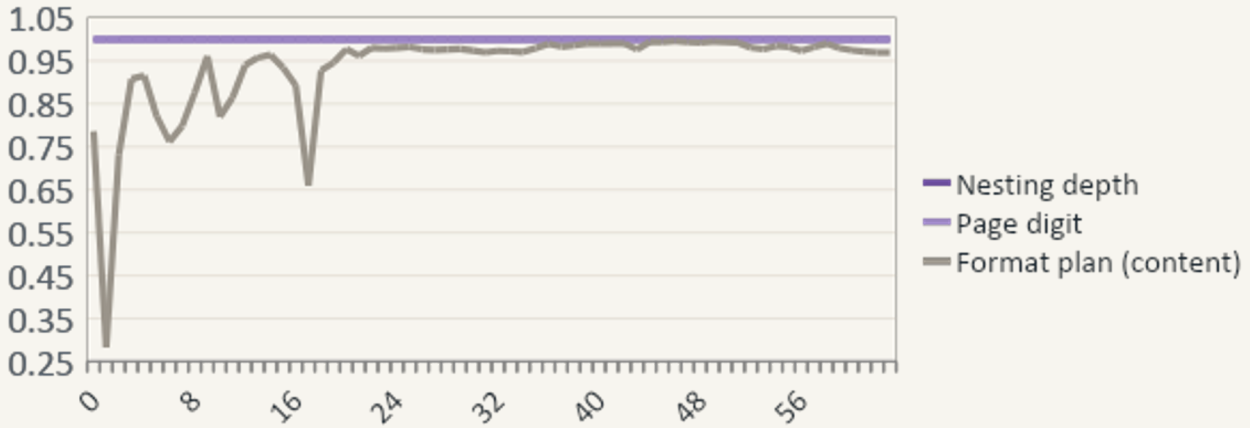
decodable causal here stated

Inside the reader the structural state is perfectly decodable, but seldom stated

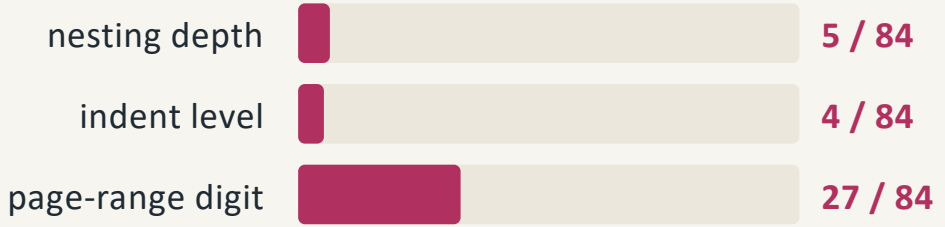
Where the probe reads



Decodable · TEST AUROC at the first generated token, reader block 0–63



Stated · in E100's readouts, the Oracle states any value of



Probe every reader block (0–63) at two positions: the injected marker and the first generated token. 84 dev / 84 test items per row; dev-only model selection. The first generated token is always the same token (ID 12), so token identity cannot explain it.

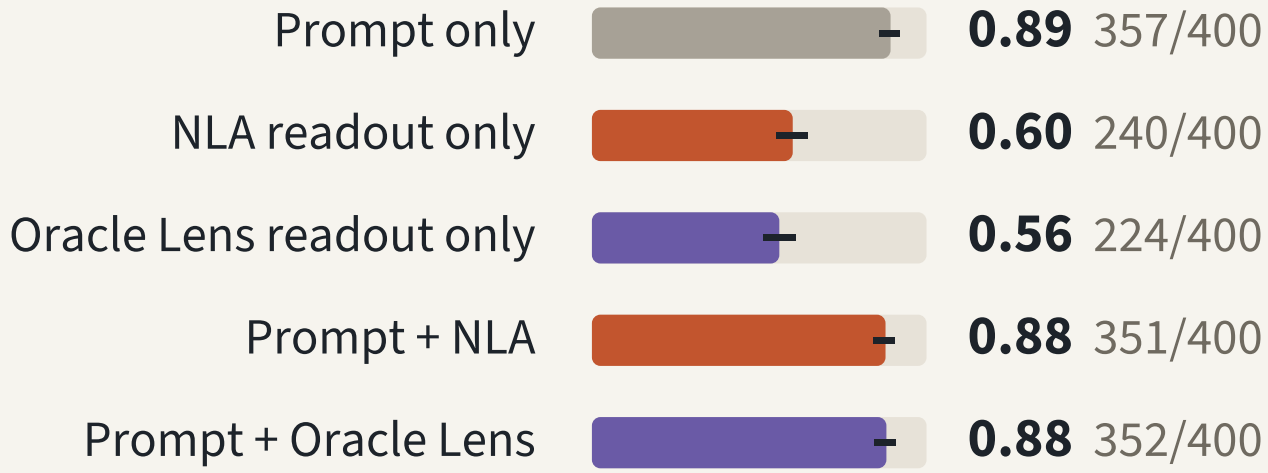
Does not show That the reader uses the state: at the injected marker the vector is its own input, so 1.00 there is trivial; the first-token probe shows the state reaches generation positions, not that it drives words.

Experiment: E113

WORKSPACEBENCH · 400 ITEMS · ONE JUDGE ANSWERS FROM DIFFERENT INPUTS

Readout added to the prompt → no gain

Judge answers correctly (share of 400 items)



Prompt (typo item)
…as in Cage's Music of Changes, is called aleotoric

NLA, from the activation
… is building toward naming the concept/term …

Same model, prompt only
… demands the standard term "aleatoric music" …

Potential · Method: 0/187 NLA passes are out of reach of prompt-only text → hold text fixed.

≈11% of items are left where reading could show. Prompt + readout − prompt: NLA −0.015 [−0.043, +0.013].
Wilson 95%. 0/187 is vs best-of-8 prompt-only texts. Card: first item passed by both. Exps 01, 06.

14 CONCEPTS · 160 ITEMS PER CELL

Asked vs unprompted → Oracle Lens is silent

		Asked prompted AUROC			Unprompted % of 320 free readouts		
		NLA	Oracle Lens	J-lens	NLA states	Oracle Lens states	Oracle Lens exhibits
position in text	line position	0.60	0.55	0.48	0%	0%	0%
	line break coming	0.54	0.55	0.45	0%	0%	0%
	words into the sentence	0.72	0.55	0.46	0%	0%	0%
	list item index	**1.00**	**0.99**	0.57	**25%**	0%	0%
	indentation level	**0.87**	**0.89**	0.68	0.3%	0%	0%
	inside a quotation	**0.83**	0.76	0.49	**14%**	0%	0%
next word	next word's part of speech	0.72	0.66	0.52	**31%**	0%	0%
	upcoming noun's gender	**0.89**	**0.92**	0.60	**9%**	0%	0%
	upcoming noun's number	**0.99**	**0.86**	**0.84**	**28%**	0%	0%
controls	tense	**0.90**	**0.92**	0.53	0.3%	0%	**16%**
	all caps or not	**0.86**	**0.93**	0.52	**7%**	0%	**50%**
	'Item N' number	**0.98**	**0.99**	0.64	**8%**	0%	0%
	French or German	**1.00**	**1.00**	**0.98**	**81%**	0.3%	**56%**
	code language	**0.99**	**0.99**	**0.95**	**82%**	0%	**93%**

Asked (prompted AUROC ≥ 0.80)
NLA on 10 of 14 concepts, Oracle Lens 9, J-lens 3.

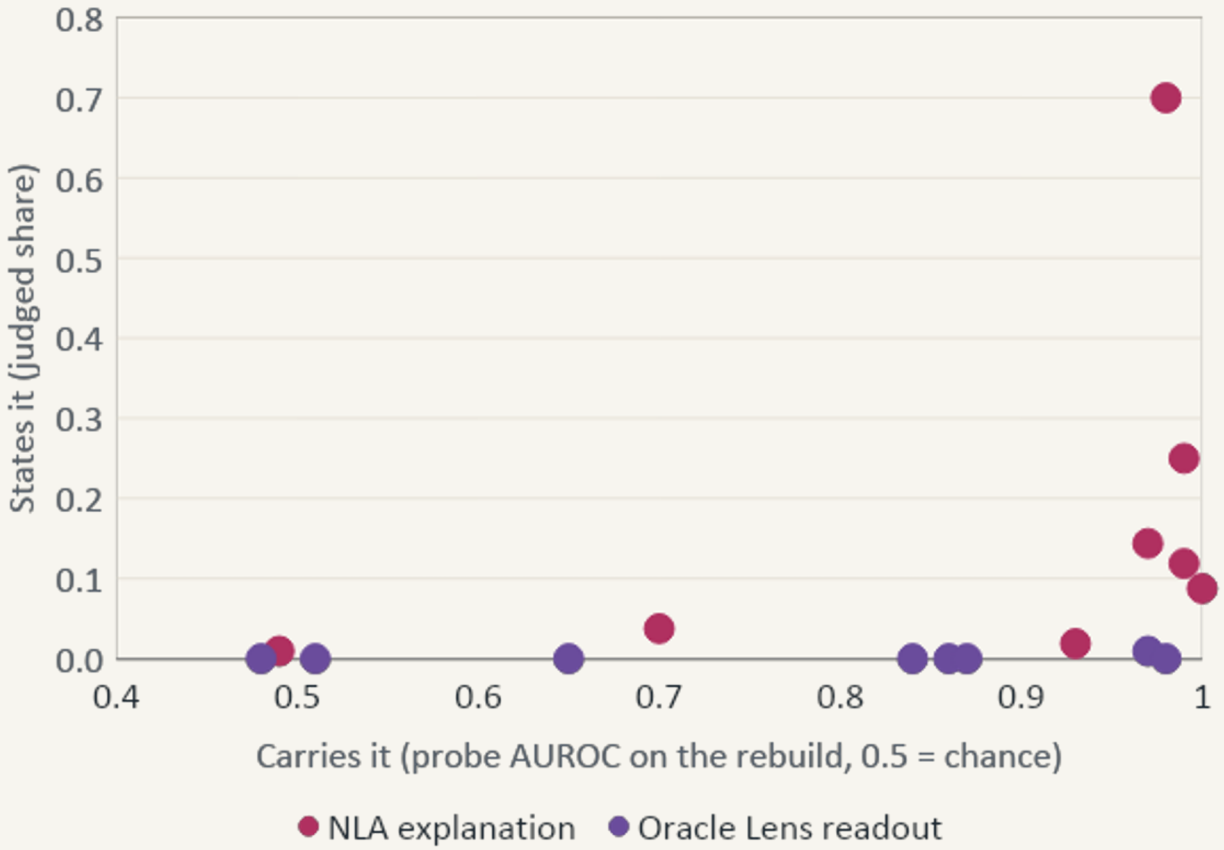
Unprompted (judge)
Oracle Lens states a concept in 1 of 4,480 readouts; it writes in language, code, caps and tense instead.

Potential · Method: position in text is readable, detected, rarely stated: the test family.

Unprompted: judge on 320 free readouts per concept. One judge (Gemini 3.8 Flash); intervals in backup.

decodable causal here stated

Readers' text often carries structural state that is rarely stated



TestCase	Carries · Oracle / NLA	States · Oracle / NLA
Python string	0.86 / 0.99	0.00 / 0.12
JS string	0.84 / 0.97	0.00 / 0.14
LaTeX math	0.87 / 1.00	0.00 / 0.09
Call argument	0.51 / 0.93	0.00 / 0.02
JSON depth	0.97 / 0.99	0.01 / 0.25
MD heading	0.65 / 0.70	0.00 / 0.04
MD list	0.48 / 0.49	0.00 / 0.01
Code language (control)	0.98 / 0.98	0.00 / 0.70

Carried, not stated: the readout reproduces the text that holds the state, and does not name it.

Does not show That the readers use the state, or that the Oracle carries less: its readouts are rebuilt by the NLA's critic, which is out of distribution for them. Carry and stated rates come from different splits of the same bank.

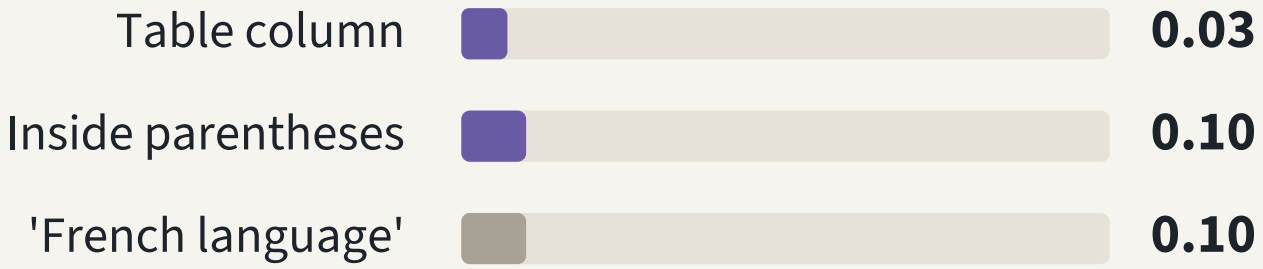
MECHANISM · EXP 14 · ORACLE LENS'S RECONSTRUCTOR · 200 ITEMS PER ROW

A statement vs real text → only text rebuilds

Rebuilt from the true 32 tokens (probe AUROC)



Moved by a one-line statement (share of class gap)



The statement pair

"- *column one*" *vs* "- *column five*"

A true span (end)

```
| Iris | Italian |
principal |…
| Farid
```

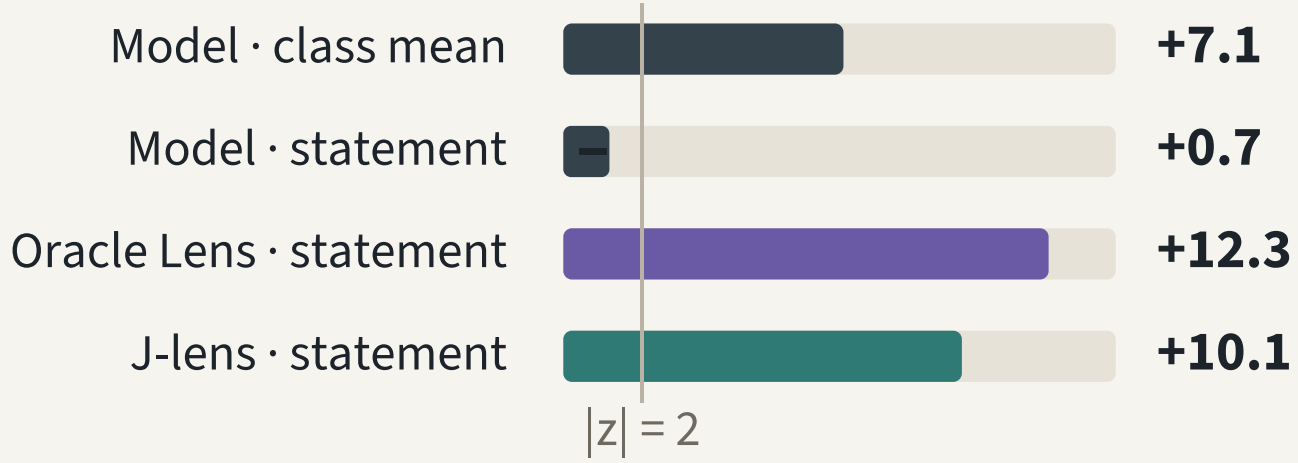
Potential · 1-(b), 2-(a): statements barely move a reconstructor; no reward for stating.

Oracle Lens's reconstructor (layer 44). Aggregates, no intervals. NLA critic, same statements: 0.07, −0.05, language 0.72.
A bullet written in French moves the rebuild 1.39. Exp 14.

MECHANISM · EXP 15 · TABLE COLUMN · 200 ITEMS · PUSH 11% OF THE NORM

Readers respond to a direction that barely affects Qwen’s next-token behavior

Response to the push (z against 20 random directions)



Statement pair the direction is built from

"The current cell is in the first column of the table" vs "... fifth column ..."

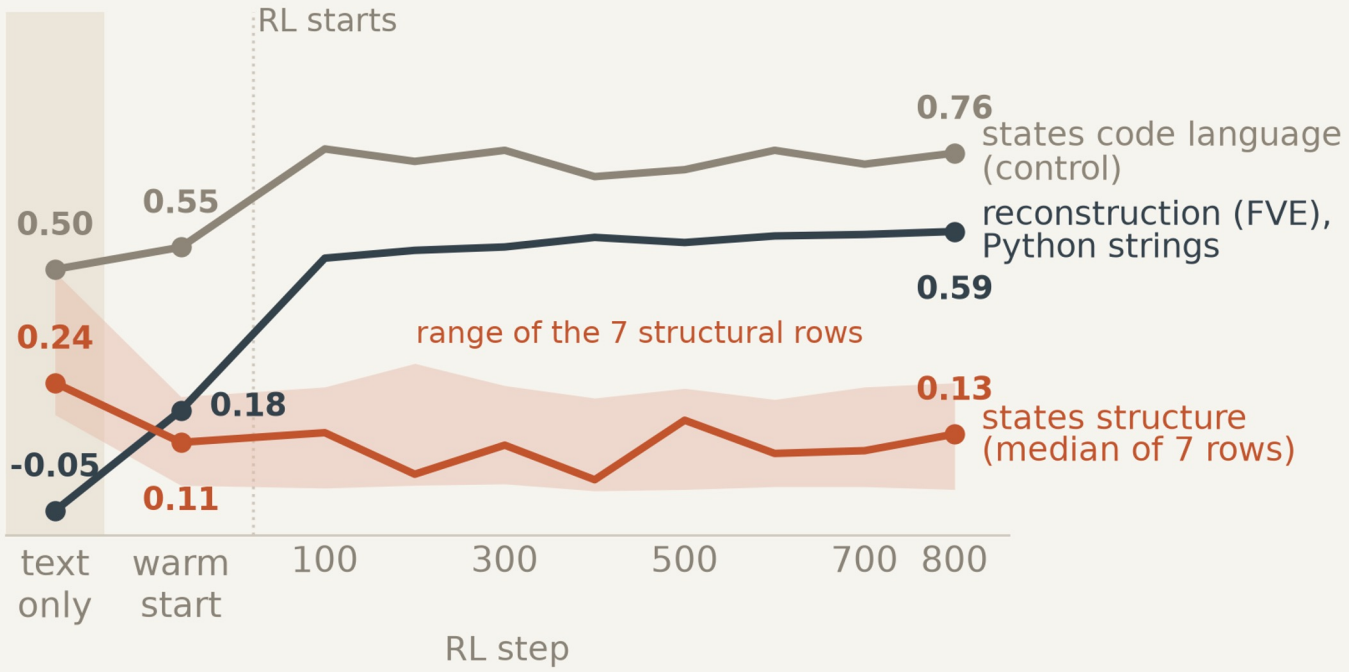
What 'model' means

The shift in the model's own next-token log-odds for a newline.

Potential · Why it matters: a reader can follow a direction the model ignores.

Directions built with the NLA's reconstructor, so the NLA's response is partly circular (not shown). Parenthesis: model −3.6, Oracle Lens +3.6. Model statement z CI [0.4, 1.1], cluster bootstrap by passage. Exps 15, 17.

More RL → better reconstruction, not stating



Input (inside a Python string)

```
raise
DMBadInputError("Shape of
rep does not
```

Text-only warm start (states it)

... requires specific error message text ...

After 800 RL steps (doesn't)

... ends "does not" mid-error-message ...

Potential · 2-(a), 2-(c): RL raised reconstruction; stating stayed at the warm start's level.

Structural rows 0.00–0.29 over RL, no trend (slope −0.001 per 100 steps). One final critic scores every checkpoint.
In-string readouts: text-only states it 142/160, step 800 59/160 (less often, not rarely). Card: first of 12 such. Exp 29.

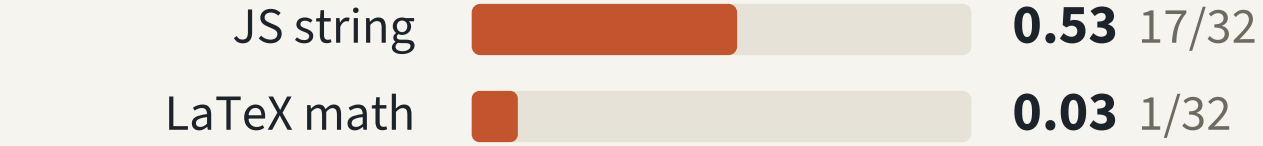
WHY · 400-STEP SUPERVISED ADAPTER · 32 HELD-OUT ITEMS PER ROW

Targeted supervision improves naming on trained properties, but generalization remains weak.

Trained rows: correct when it names



Untrained rows: names the state at all



Adapter NLA, its last line

The current position is inside a string literal

Base NLA, same item

... is mid-sentence inside a quoted error string ...

Reconstruction (FVE)

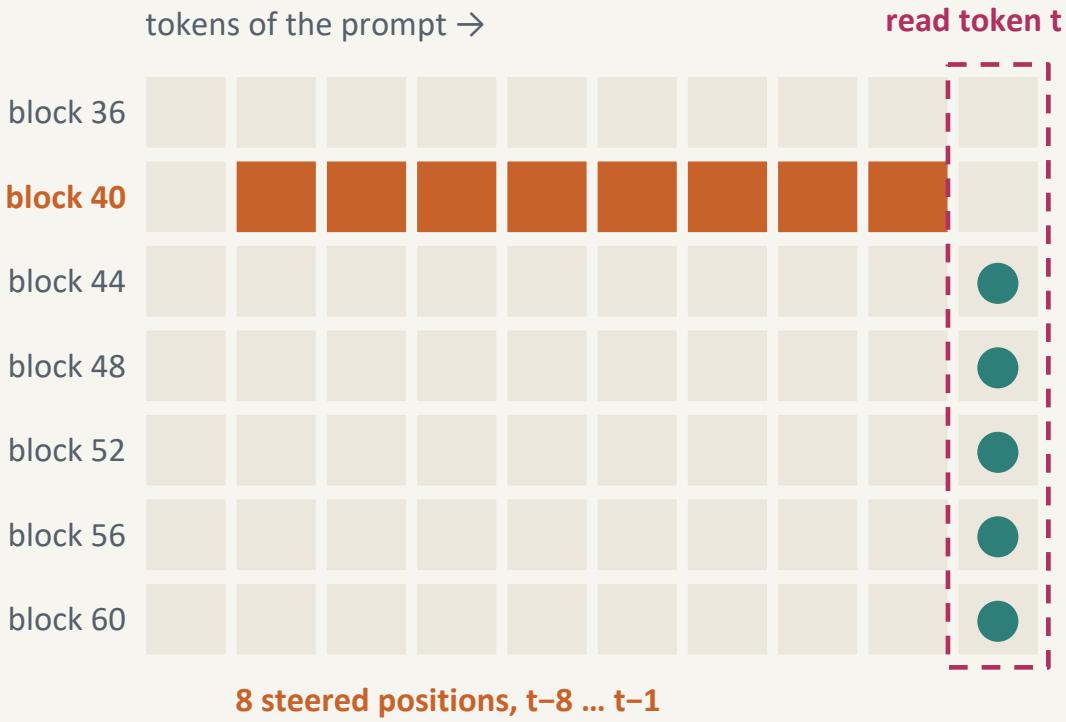
Adapter 0.507 vs control 0.512.

Potential · 2-(b): naming isn't blocked by capacity, on rows it was trained on.

Base NLA's 0/512 is a substring test for the trained sentence; it already says 'inside a quoted error string'. Wilson 95%.
Markdown is barely readable; list depth 11/24. 3 of 5 predictions failed. Card: first correct item. Exp 30.

decodable · causal here · stated

Push the answer upstream at block 40, then put one clean token back at a later block



steer: block 40, the 8 positions before t; t itself untouched

restore: clean activation into t at one block ℓ, one run per ℓ

1 Steer

$$\tilde{h}_{40,i} = h_{40,i} + \alpha \cdot \|d\| \cdot \hat{d} \quad i = t-8 \ldots t-1$$

d: class-mean difference of block-40 activations on 84 dev items · at α = 8 the push is 2.23× the median block-40 norm

2 Score Qwen's answer

$$\Delta = \log p(\text{target continuation}) - \log p(\text{original})$$

fixed continuations: the steering target vs Qwen's original answer

3 Restore one cell, measure what is undone

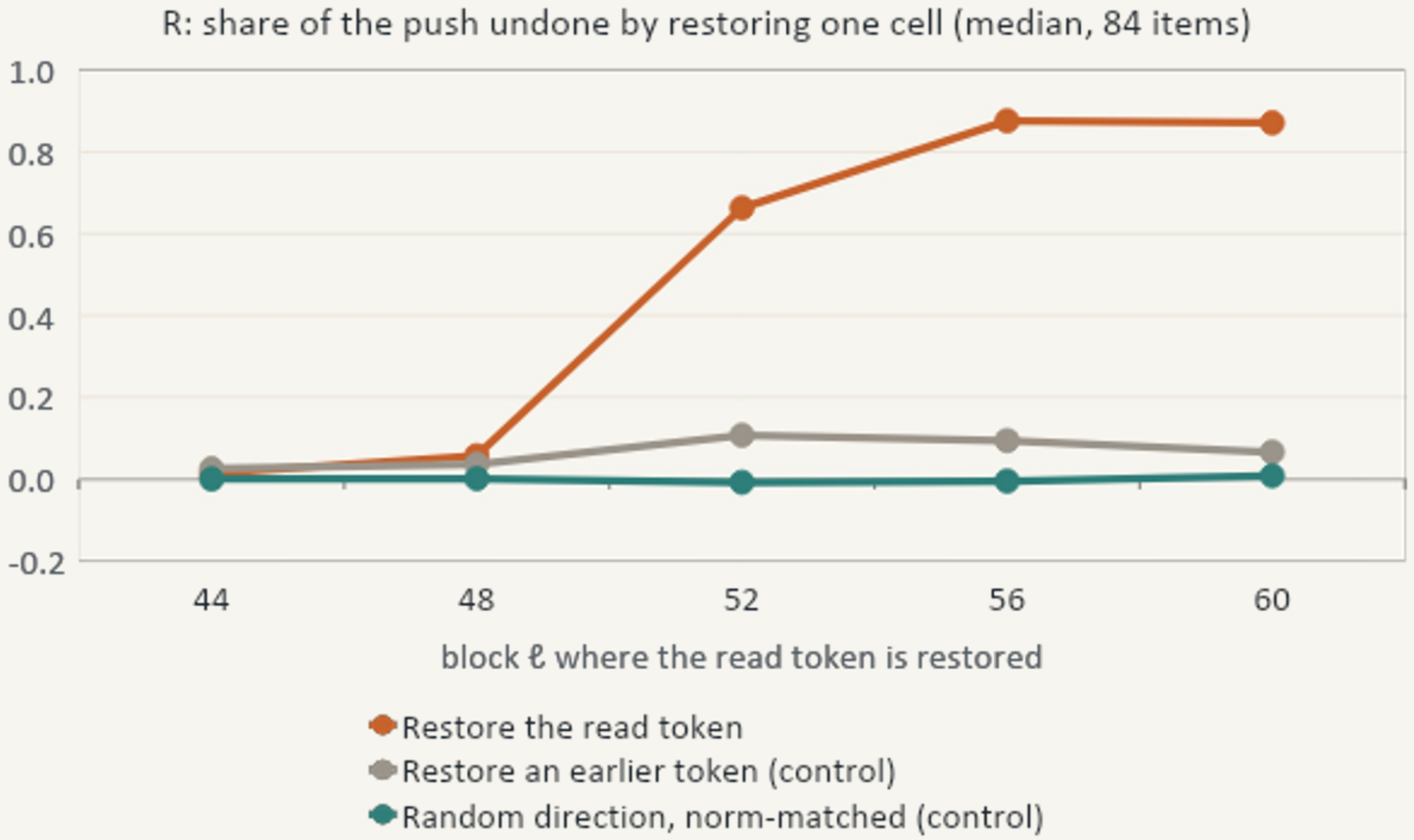
$$R_\ell = (\Delta_{\text{steered}} - \Delta_{\text{restored},\ell}) \,/\, (\Delta_{\text{steered}} - \Delta_{\text{clean}})$$

0: nothing undone · 1: the whole push undone · median over 84 items

Does not show A natural perturbation: the push is 2.23× the median activation norm, and a random direction of that size flips 42 of 84 pair preferences (the real one 84 of 84). Restoration measures sensitivity at one token, not when the model first "knows".

decodable causal here stated

Restoring the read token becomes strongly effective between blocks 48 and 52



Two different outcome measures

84 / 84

Qwen's pair logit prefers the new answer under the push (E104)

43 / 84

Qwen's own free text states the new answer; 41 state neither (E131, NOT MET)

S −0.52 at 44, +1.00 at 52

Oracle reader: still the old answer at block 44; the new one on 84/84 at 52 and 60 (E109)

Does not show Generality: one bank under a 2.2× push. The bank's language row is non-monotone (R 0.29 already at block 44). R is a median over 84 items (52: 0.66 [0.51, 0.80]), not when the model first computes the answer.

decodable causal here stated

Unsteered transfer chains: swap only the final " by" token with its opposite twin

Twin A

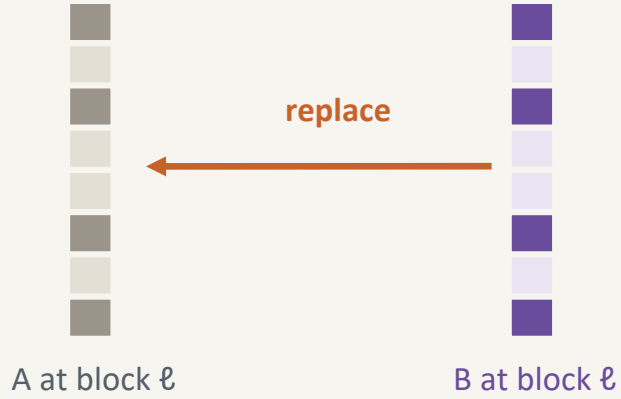
```
The brass key starts with Alice. Alice hands the brass key
to Bob. Bob hands the brass key to Dana. Dana hands the
brass key to Erin. Later, Carl asks Dana about the brass
key. Bob waves to Carl. The brass key is now held by
```

Twin B · differs only in the last recipient

```
… Bob hands the brass key to Dana. Dana hands the brass key
to Frank. Later, Carl asks Dana about the brass key. Bob
waves to Carl. The brass key is now held by
```

Anti-recency: one or two sentences after the last hand-over mention other names, so the most recent name is never the holder. Text assembled from the bank's templates for illustration; sealed items are not in the repo.

Swap at the raw final token " by"



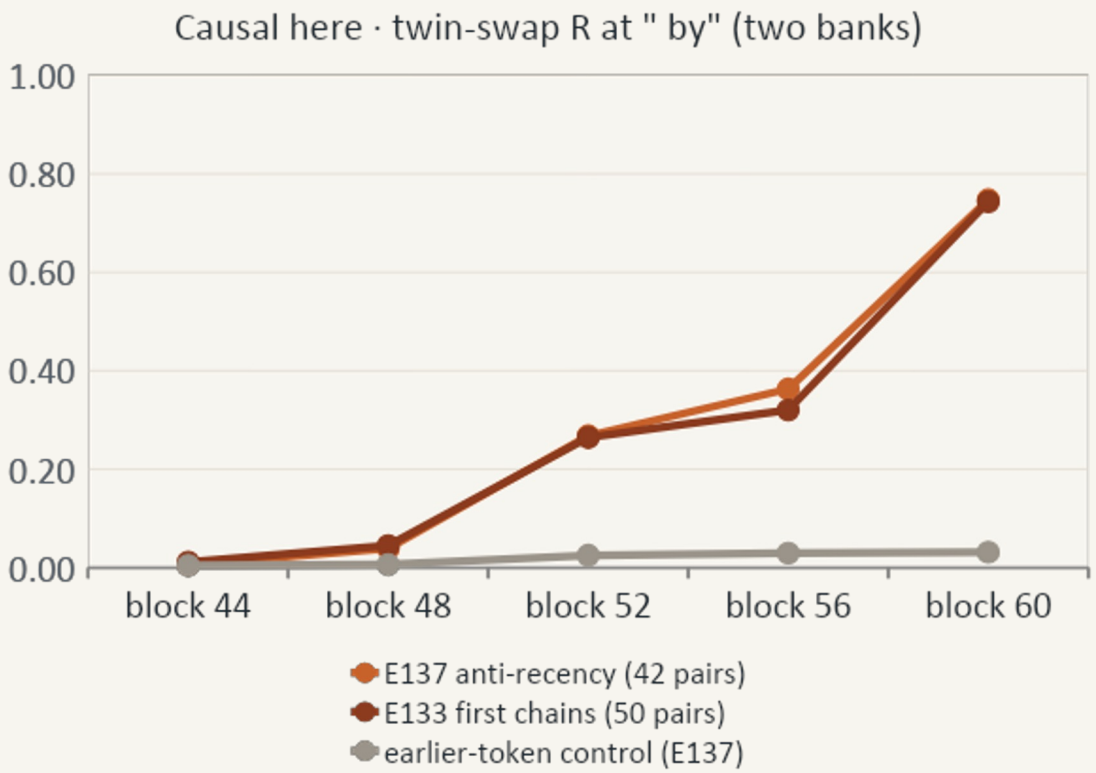
$$R_\ell = (\Delta_{swap} - \Delta_{own}) / (\Delta_{twin} - \Delta_{own})$$

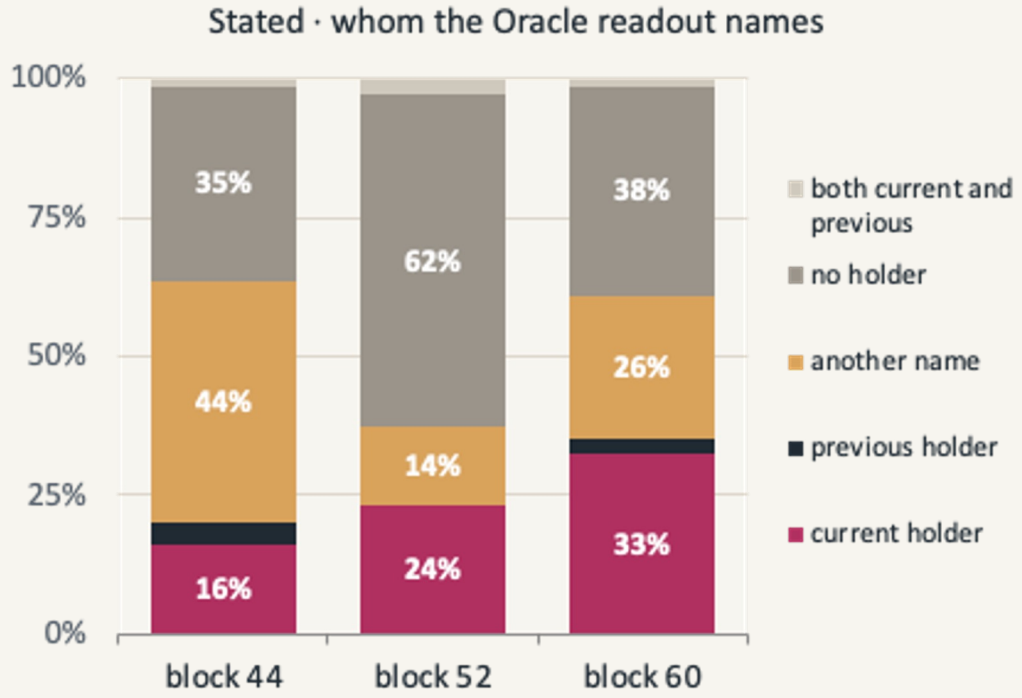
Δ: Qwen's logit difference between the two recipients. 0 = no effect · 1 = behaves like the twin.

Here nothing is steered: the read token's activation is replaced by the twin's, whose answer differs.

decodable causal here stated

Unsteered chains: the final token matters mostly at block 60, and the previous holder is rarely named





Decodable? Mixed: macro AUROC 0.978 at 44 over 8 of 10 holder classes; the registered twin probe is at chance (≈ 0.43)

Does not show Timing: the registered later-site rule (R ≥ 0.50 at 52 or 56) was NOT MET, and staleness (previous holder ≥ 0.30 at 44) was 3 of 79. Synthetic chains; 5–8 of 84 readouts missing per block.

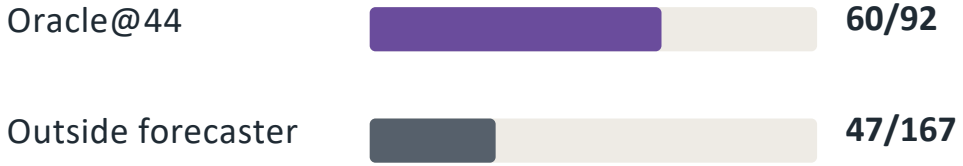
E137 (confirmatory, natural-staleness criterion R-NL2 NOT MET) · guivff/activation-readers @ c60fa970 · e137/analysis/sealed_main/RESULTS.json (design D1, 42 pairs) · bank README_BANK.md · reviewed R-4

decodable causal here stated

Some readouts align with subject-specific mistakes better than outside forecasters

E107b · NATURALLY WRONG ANSWERS

Conditional on a value stated:



All-item rates: 60/187 vs 47/196.
Coverage / model-family mismatch matters.

E116 · NON-MODAL ANSWERS (POST HOC)

Matched 110-item subset:



Paired advantage +25.5 pp
95% CI [+11.8, +38.2] pp

E131 §246 (post hoc): Qwen generated the new answer in 43 cases; early Oracle@44 still said old in 26/43, while @52 said new in 43/43.

Does not show Matched model-private access: E107b has unmatched denominators and E116 is post hoc on a selected cohort.

What we got wrong—and why the remaining results might be boring

FAILURES THAT CHANGED OUR PRACTICE

1. Judging inputs, not just outputs
E138 recipient answer split into characters.

2. Trivial baselines before intricate rules
S19: always-low 33/38 vs our 31/38.

3. Generated behavior instead of logits
E131: 84/84 pair preferences, only 43/84 generated target.

REASONS THE STORY COULD BE BORING

1. One shared backbone and text-explainable tasks
Benchmark successes may not use private state.

2. Many contexts, but few unique facts
The strongest steered fact result used one fact pair.

3. Natural chains retain a shortcut
The last successful transfer still identifies the holder.

DO DIFFERENTLY: pre-register the trivial baseline; human-code a blind sample; test several fact pairs and negated transfers.

decodable causal here stated

Three explanations survive: the decisive test must separate them

1 · PROMPT RECONSTRUCTION

Hypothesis: the reader infers the question and uses its own prior.

Test: fixed text, verified different subject states, matched text-only predictor.

2 · OUTPUT REGISTER

Hypothesis: state is accessible but normal generation rarely expresses it.

Test: fixed activation, controlled output elicitation; human-validated naming.

3 · TRAINING OBJECTIVE

Hypothesis: reconstruction optimization fails to reward truthful statements.

Test: matched warm start, reward-only ablation and held-out generalization.

Does not show That one positive probe, donor swap, reconstruction score, or readout is sufficient to identify a mechanism.

RESEARCH PHASE · FIRST PRIORITY

A benchmark of model-private state, verified causally, against strong text-only baselines

Manipulation

Items where Qwen's answer is not recoverable from the text alone: wording-dependent answers (58 of 300 screened in E140), facts Qwen gets consistently wrong, two-step questions through a wrong intermediate.

Verify each item's state twice: repeated sampling of Qwen, and a read-token patch that moves the answer.

Arms and matched baselines

Reader on the activation (Oracle, NLA; several blocks)

Same backbone given only the text the reader decodes

Same backbone given the prompt, k-matched samples

Linear probe on the answer logit

Endpoint

Tracking rate: the readout states Qwen's own per-item answer, crossed with question identity (same / different) × verified state (same / different).

Powered for a 0.15 difference over the best baseline at matched coverage; missing readouts counted as failures.

Falsifier The readout follows question identity rather than verified state, or a powered interval excludes a useful edge over the best text-only baseline. Mining is the cost: the 58 discordant pairs came from 300 screened.

RESEARCH PHASE · SECOND, ONCE THE BENCHMARK EXISTS

Pay the reader for true statements the prompt text cannot predict, and measure false claims

Manipulation

Step 0, no training: score true, false, neutral and exhibiting statements under the exact NLA training reward (E139 only had a proxy).

Then train from one warm start, ≥ 3 seeds per arm, with the reward as the only difference between arms.

Arms and matched baselines

Reconstruction only (current objective)

+ reward for statements a held-out probe or patch confirms

Reconstruction scored only on what a text-only model misses

Supervised statements as a reference

Endpoint

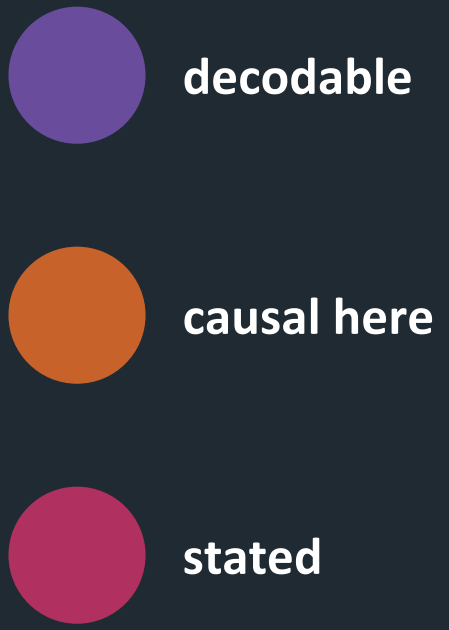
On held-out properties and read positions: correct statements, false statements and FVE, reported together.

Success: naming rises ≥ 0.10 with learning confirmed, false claims rise less than correct ones, FVE within 0.02 of control.

Falsifier Naming does not rise with learning confirmed, or false claims rise as fast as correct ones (the exp 30 random-position pattern), or FVE drops.

CLOSING

Decodable, causal here, and stated are different. A reader faithful to the text is not yet faithful to the model.



What would convince us a reader reports model-private state: a powered benchmark of verified private state, read causally, where the reader beats a same-backbone model given only the text.

We have built the parts of that test that failed for us. The next two experiments are designed to fail too, if we are wrong.

decodable causal here stated

Appendix map, frozen sources and who checked what

Slide	Contents	Slide	Contents
A2	Registered rules that failed, and what we now say	A9	Exp 29: every row, every checkpoint
A3	Read sites and injection, per experiment	A10	Exp 30: registered P1–P5
A4	Three interventions, three questions (E138R, E109, E140)	A11	E139: the proxy reward for true vs false
A5	E100 and E113 in detail	A12	Steering bank: dose, controls, free text (E104–E135)
A6	The cue experiments (E85, E120, E136)	A13	Transfer chains and cross-depth table (E133, E137, E95)
A7	Suraj exp 06 in detail	A14	The human audit (E71) and the E138 timeline
A8	S16 vs E7, and exp 28's withdrawn rule	A15	Cheap monitoring checks (E121, E118, E123, E134)

Note No file in the repository tallies registrations by verdict, so we do not quote one. "139" counts distinct experiment numbers in the brief (E1–E140 without E55), not registrations.

Hypotheses

> *Both readers carry structural state in what they write. Oracle Lens never states it; the NLA states it in about one readout in four, often loosely, and that rate looks set by its warm start.*

H1 · 85%	**Readers carry structural state but don't state it.**
1-(a).	The state is recoverable from what they write.
1-(b).	Oracle Lens never states it.
1-(c).	The NLA states it in a minority of readouts (about 1 in 4), often loosely.
H2 · 55%	**The NLA's silence is a warm-start habit.**
2-(a).	RL's reward doesn't pay for structural statements.
2-(b).	Naming isn't blocked by capacity.
2-(c).	The habit is set before RL.

SETUP · THREE READERS

One activation, three readers, no prompt

NLA
Activation added to its prompt; writes an explanation.

Oracle Lens
Activation as one character's embedding; writes bullets.

J-lens
Projects onto the J-space; lists top tokens.

Same activation, Python source, read at ▌: `"is not supported. Support for▌`

NLA
... is mid-sentence, requiring completion of this standard ONNX export error pattern ...

Oracle Lens
某些 operators is in progress. Please open an issue on GitHub to request support

J-lens, top tokens
この · implementation · эмого · implementations · Generic ...

NLA: ceselder iter_000400. Oracle Lens: agu18dec rl600 (trained at layers 20–60, step 4). J-lens: n1000. Layer 42.
Item: first in exp 22 where neither reader's first readout uses a string word (5 of 20; the NLA usually does).

SETUP · FOUR MEASURES

Four measures per state, fixed in advance

Readable
A probe on the activation separates the two values: AUROC ≥ 0.80, folds grouped by source file.

Carries
The reader's text brings the state back: the critic rebuilds the activation from it and a frozen probe scores it, or a word-count classifier reads it.

States
A blind judge (Gemini 3.8 Flash) reads only the readout and says whether it names the value. 'Names' means a strict regex.

Used
Push the activation 11% of its norm along the state's direction; the model's next-token shift is a z against 20 random directions.

Banks: synthetic (exps 12–18), natural (exp 19, withheld, hashed), public (exp 22), random points (exps 23, 27). Predictions written first; hashed for exps 18, 19, 22.

New relative to the papers: they score what readers say; we ask what their words carry unsaid.

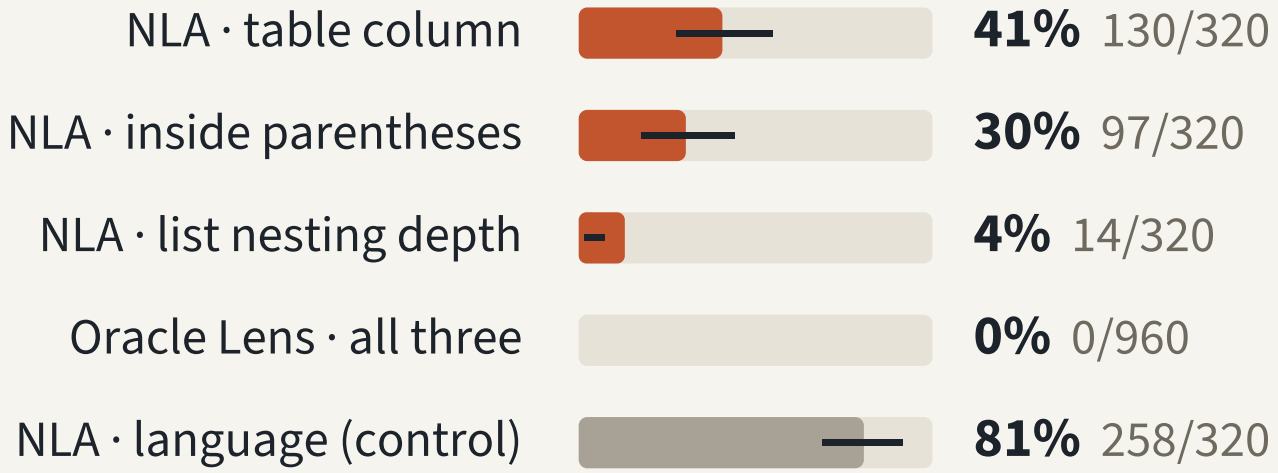
The objects and instruments behind every number

Qwen3.6-27B, layer 42	Subject model. Every read is the residual activation at one token, layer 42; the NLA is hard-wired to it.
NLA, ceselder iter_000400	Actor writes an explanation of the injected vector; its critic rebuilds the vector from the words; RL pays for the rebuild.
Oracle Lens, agu18dec rl600	LoRA on the same model, trained at layers 20–60 in steps of 4, so 42 is off its grid. Vector goes in one slot character; writes bullets.
J-lens, n1000	Projects the vector onto the rank-1,536 J-space at layer 42 and lists the top tokens. Generates no text.
WorkspaceBench	The benchmark these readers were built against. Regex-scored; four families, 400 items in our runs.
Our instruments	Readable: probe AUROC ≥ 0.80. Sayable: critic rebuild from 32 real tokens ≥ 0.80. Used: 11% push, $\|z\| \geq 3$.

POSITION IN TEXT · FREE READOUTS, 40 ITEMS × 8 PER ROW

Unprompted → only the NLA states position

States the value (judge, share of 320 readouts)



Input (first cell of a row)

```
| Iris | Italian |…
| Farid
```

NLA (judge: states it)

… *"Farid" is the start of a new name entry* …

Oracle Lens, first item (writes cells)

/Farid | France | Asie | 32 | 10 | Informatique |

NLA, rare literal case

… *it is the first column of the next row* …

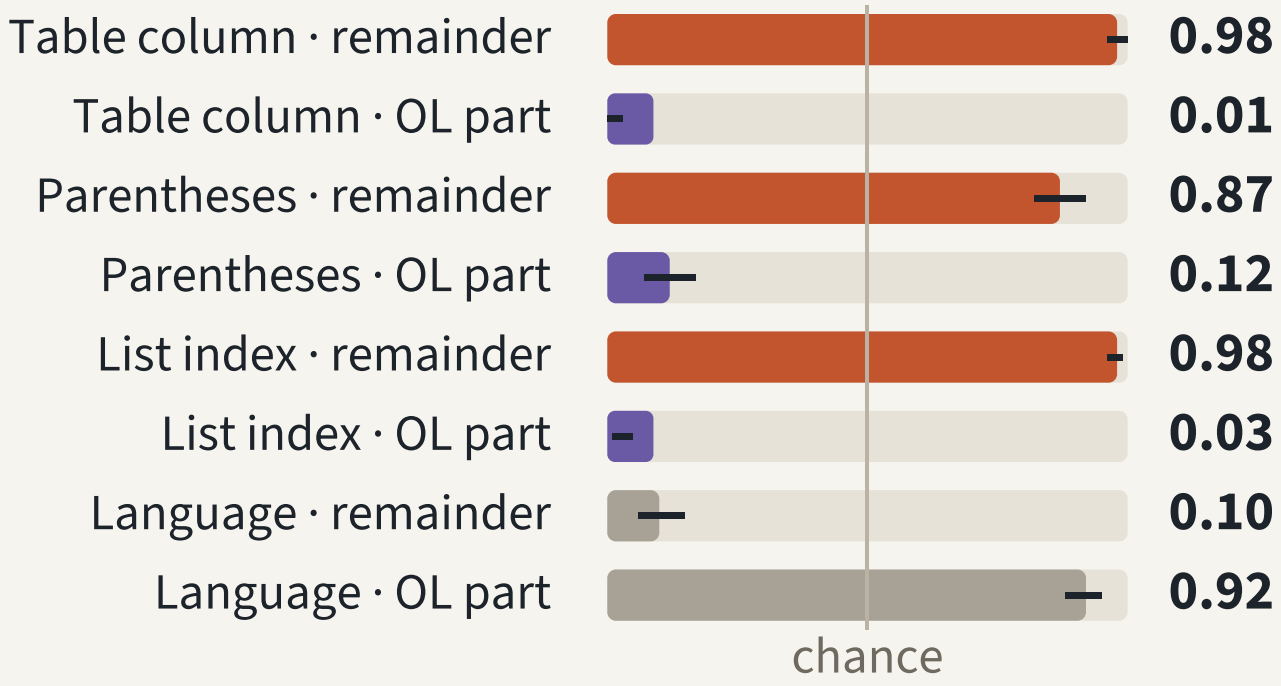
Potential · 1-(b), 1-(c): Oracle Lens states none; the NLA states two of three, loosely.

Judge counts 'a new row is starting' as stating; 7 of 118 judged statements say 'first column'. Item-clustered 95%.
Line position on fresh items: chance for every reader (0.51, 0.49, 0.53). One judge.

MECHANISM · 200 ITEMS PER ROW

Swap Oracle Lens's part → the NLA ignores it

NLA follows the donor (AUROC against the donor's label)



OL part
What the NLA's reconstructor rebuilds from Oracle Lens's bullets.

Remainder
The activation minus that part. Swap one part in from a donor of the other class.

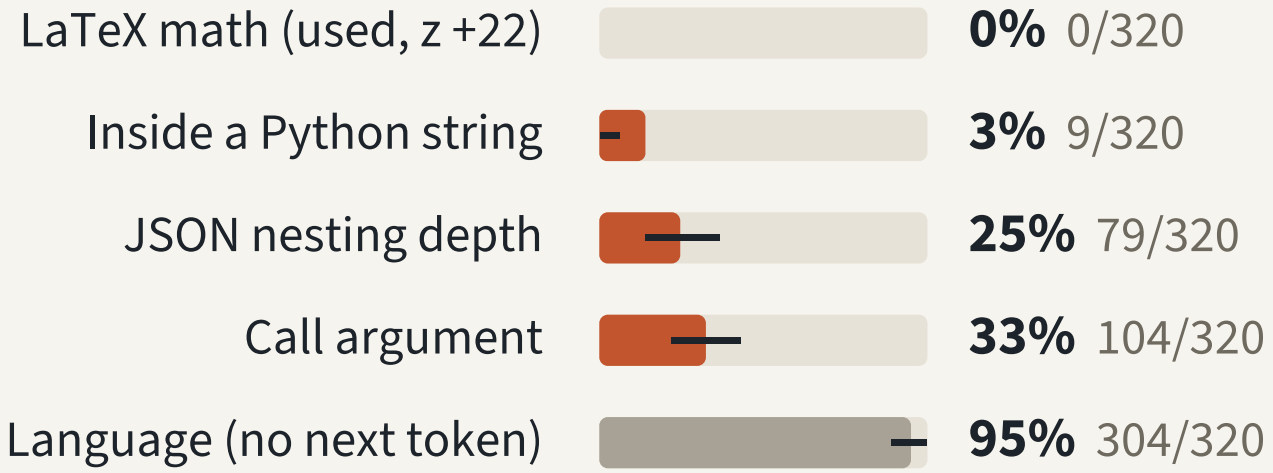
Potential · 1-(b): position sits in the part Oracle Lens's words don't rebuild.

The NLA's prompted log-odds, mean of 4 phrasings; 95% bootstrap CI by passage. Both parts built with the NLA's reconstructor; one seeded donor pairing.

NEGATIVE RESULT 1 · 10 SYNTHETIC ROWS · RULE HASHED BEFORE READERS RAN

Used by the model → not more often stated

NLA states it (judge, share of 320 readouts)



Pre-registered rule

"The NLA states a property exactly when the model uses it."

Scored

5 of 10 rows. Whole scorecard 44 of 70 cells against a declared 0.80; always predicting 'low' gets 40 of 70.

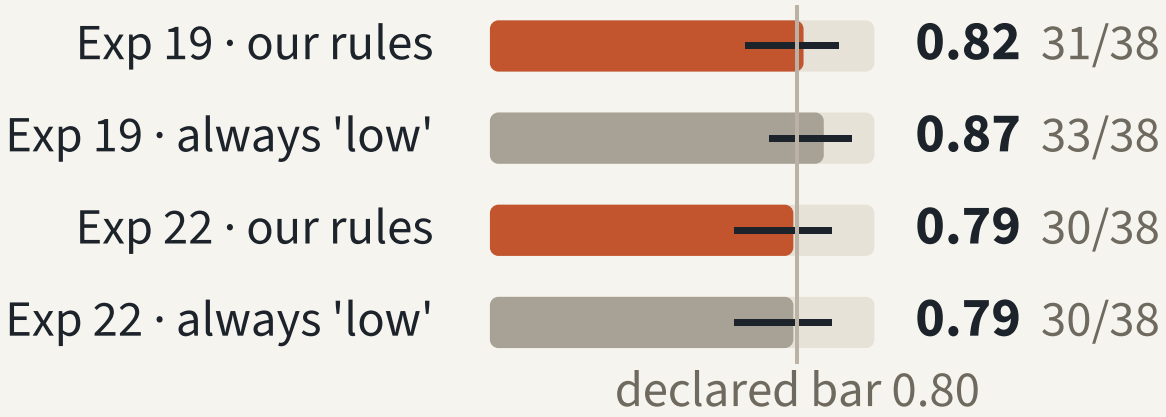
Potential · Lesson: the NLA states what text can say, not what the model uses.

Used = |z| of the push on the model's next token vs 20 random directions. Item-clustered bootstrap 95%.
Flaw: the synthetic banks carried vocabulary cues the J-lens read. Hashed before readers ran.

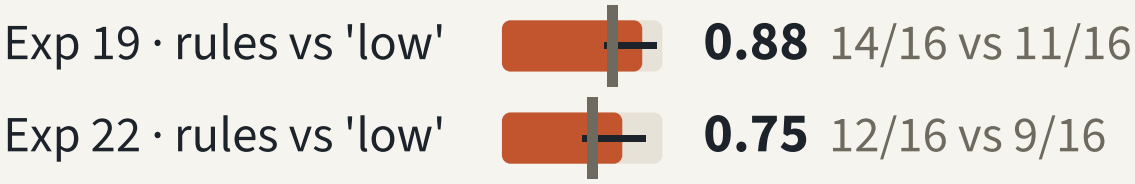
NEGATIVE RESULT 2 · PRIMARY SCORECARD (FREE READOUTS)

Always predicting 'low' → ties our rules

Primary scorecard, cells correct



Revised NLA rules only (16 cells)



When we found it
30 Oct, recomputing every figure from raw files, after we had quoted these scorecards as passing.

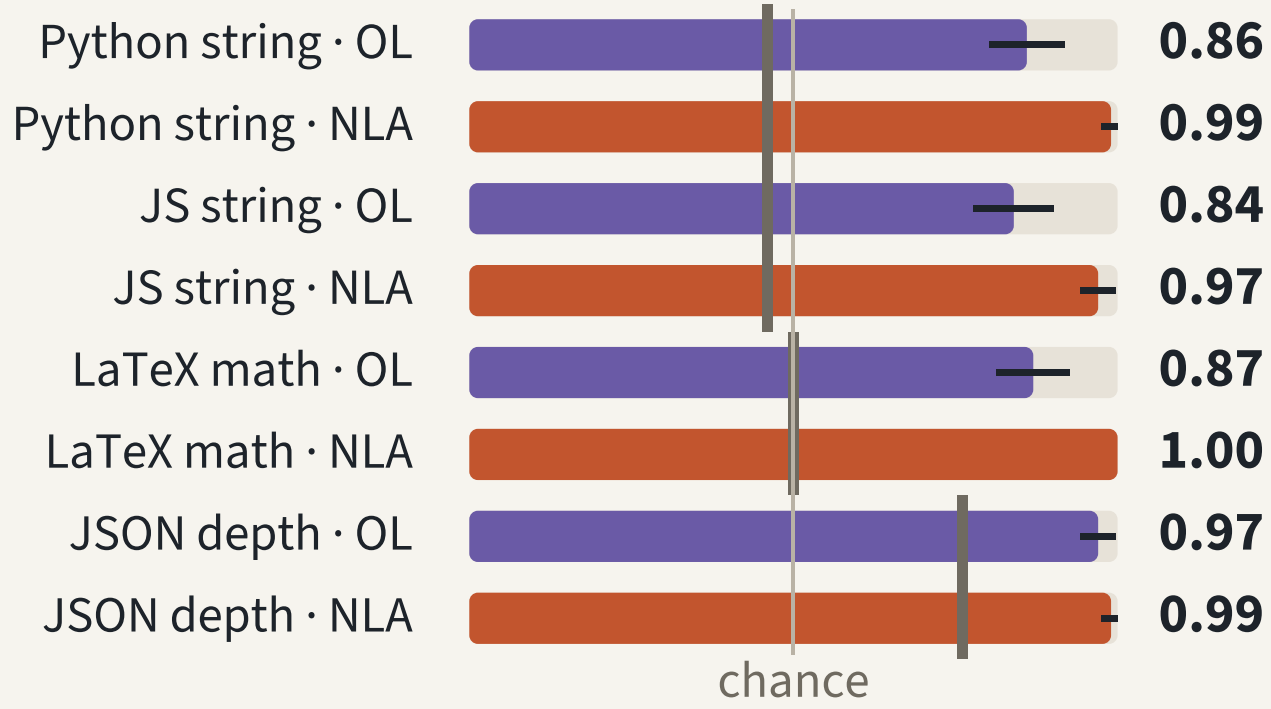
Potential · Lesson: pre-register the baseline too; only complement naming beats 'low'.

Grey tick = always 'low' on the same cells. Wilson 95% over cells. All cells: 44/56 vs 45/56 (exp 19), 44/56 vs 36/56 (exp 22).
Margin = complement naming. Exp 22 shares 229 of 1,461 source clusters (72 identical items) with exp 19: a partial replication.

CARRYING · PUBLIC BANK · 160 ITEMS PER ROW

Decode the readers' words → state comes back

Probe on the critic's rebuild of the readout



Input (Python string)

```
….debug("Exception
```

Oracle Lens (never says 'string')

/Traceback: ", file=sys.stderr)
print("Exception

Python string row

Readout 0.86; true 32 tokens 0.84; shuffled 0.46.

Potential · 1-(a): both readers' words carry the state; Oracle Lens never states it.

Grey tick = the 32 tokens shuffled. Hanley-McNeil 95%. The critic is the NLA's own (NLA bars in distribution).
Call argument: OL 0.51, NLA 0.93. Card: first in-string item whose readout has no string word (74 of 80). Exp 25.

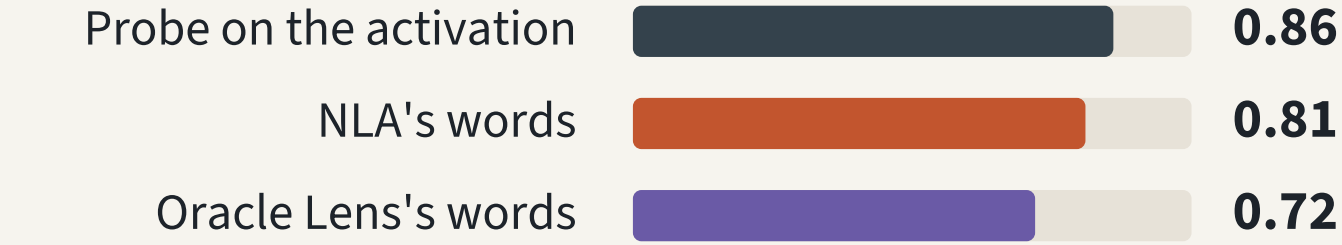
RANDOM READ POINTS · PUBLIC DOCUMENTS · NOTHING PROMPTED

Random string points → Oracle Lens is silent

States 'inside a string' (judge)



Carries it: code points, in vs out (AUROC)



Input (random point, JS file)

```
…(exports, "Valid
```

NLA (doesn't state it)

… `Valid` is the beginning of a truncated type name …

Oracle Lens

atedError | Ajv.ValidationError } from "ajv"; export type Valid

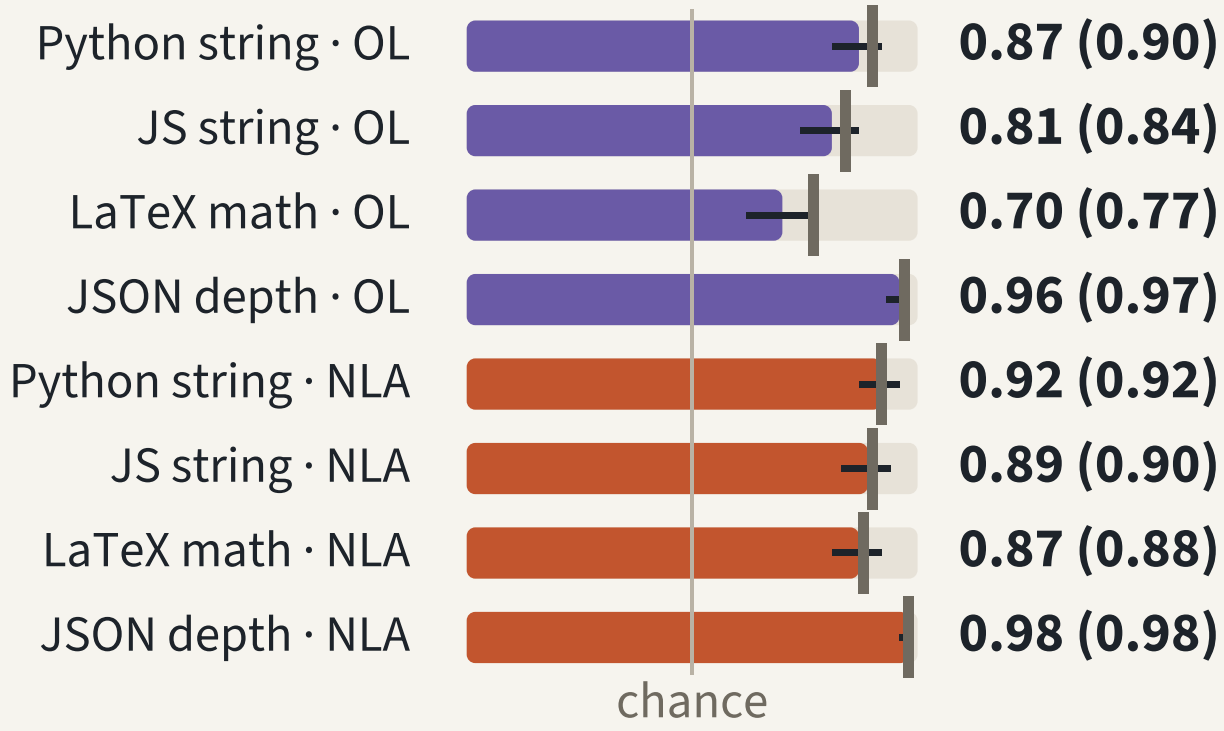
Potential · 1-(b), 1-(c): Oracle Lens 0 of 598; the NLA 22%, on code points 35 of 102.

130/602 is mostly JSON (384 of 486 in-string points). The word 'string' isn't stating: NLA uses it on 114/160 in-string and 82/160 outside readouts (exp 22). Judge: Gemini 3.8 Flash, no human labels. Card: seed-0 draw, non-stating code points.

COPY CHECK · PUBLIC BANK · 160 ITEMS PER ROW

Delete copied words → carrying barely moves

Word-count classifier, copied words deleted (AUROC)



Do readers copy the window?
Some single words: 0.19–0.24 of Oracle Lens's content tokens are in the window (0.02–0.08 by chance), NLA about half that.

Phrases?
Almost never: about 0.01 of 3-word sequences.

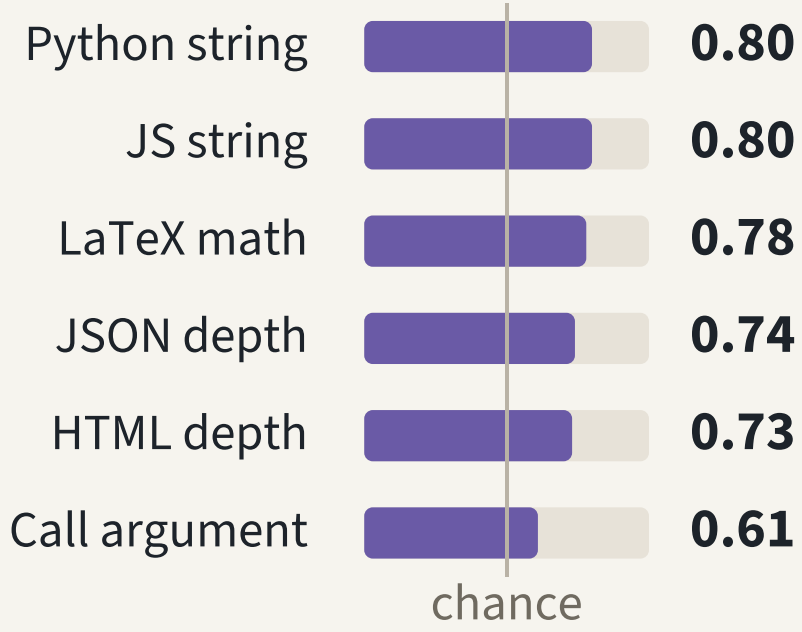
Potential · 1-(a): rules out 'it's just copying the window' for the word-level evidence.

Tick = before deletion. Leak-balanced deletion; 95% CI by source file. The window's own words: 0.72 / 0.56 / 0.62 / 0.99.
Critic AUROCs not recomputed on deleted readouts (needs a GPU). Exp 31.

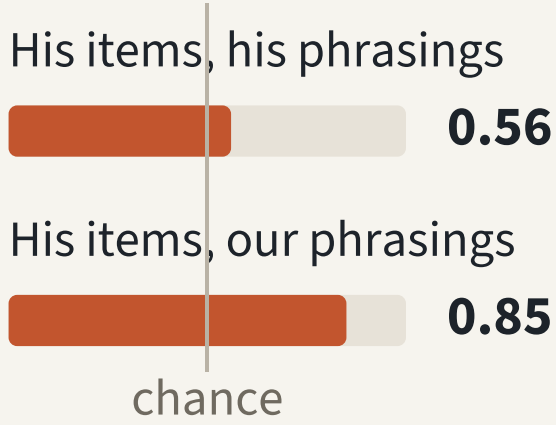
ROBUSTNESS

Another decoder, bank, layer → same picture

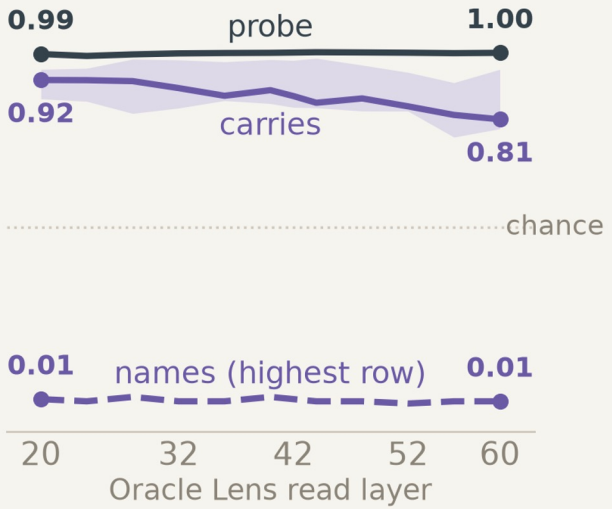
Second decoder: Oracle Lens's reconstructor (AUROC)



Guiv's bank: NLA, table column, asked (AUROC)



Oracle Lens at layers 20–60



Potential · 1-(a), 1-(b): Oracle Lens's words carry it through a new decoder, at every layer.

Second decoder: NLA explanations at chance on its frozen probe (0.44–0.56), 0.81–0.90 refit. Layers: median of 4 rows, band = min–max, no CIs. Guiv's bank also finds Oracle Lens never states the state. Exps 24, 17, 26.

Six ways this could be boring

Boring explanation	Status	What we have
Carrying is copying the last 32 tokens	Tested (exp 31)	Ruled out for the word-level evidence; critic scores not recomputed
The judge is blind to structure	Partly	Sees language 0.81–0.95 and code language 0.78; no human labels
One NLA run, one layer, one 27B model	Open	Oracle Lens flat over layers 20–60; NLA only at 42; no second model
The warm-start story is a replay	Partly	Flat over 800 steps; no counterfactual run; SFT targets never counted
Position is trivially everywhere	Tested	Shuffled tokens fall to chance; line position and Markdown unreadable
Prediction rules ≈ always 'low'	Tested: true	Always 'low' ties or beats the rules on the primary cells

PROCESS

How we kept LLM-driven research honest, and where it slipped

Hashed pre-registrations
Banks, instruments and predictions hashed before any reader ran (exps 18, 19, 22); later runs wrote predictions first.

Falsifiers that fired
Exp 18 missed its bar; we withdrew the 0.30 rule and the window account, each logged with a date.

Adversarial panel of five (D49)
Skeptic, theory and generality reviews found the confound between reward and warm start; exps 28 to 30 came from it.

Where it slipped (D51)
Recomputing every figure from raw files, found 49 written numbers that disagreed (about 23 distinct fixes); two changed a conclusion.

What we learned

1. Know what a benchmark holds fixed before trusting its score: WorkspaceBench let readers re-solve the prompt.
2. Pre-register the baseline with the prediction: always 'low' matched our rules on the primary cells.
3. Agent-written research drifts from its own data: a raw recompute found 49 wrong numbers. Audit weekly.
4. A failed prediction taught us most: exp 18's miss gave us 'states what is sayable, not what is used'.
5. A reconstructor barely sees a description, so whether a reader names a state looks like training register, not capacity.

decodable causal here stated

What we registered, what failed, and what we now say instead

Registered rule	Result	Status	What we now say
Writing the value into the prompt makes the reader state it (E114, descriptive)	0 of 3 rows	**not met**	the kind of state matters, not its format
Steering moves the block-44 reader's score by ≥ 0.30 (E104 R-U1)	+0.18 [0.08, 0.27]	**NOT MET**	a 2.2× push is not read at 44
Qwen's free text states the new fact ≥ 80% (E131 R-TX1)	43 / 84	**NOT MET**	pair preference 84/84 ≠ saying it
Unsteered chains: previous holder named ≥ 0.30 at 44 (E133, E137)	3/82, 3/79	**NOT MET**	not stale; late and rarely named
Reader follows Qwen's per-wording answer ≥ 0.60 (E140)	1 / 12	**NOT MET**	answers the question it rebuilds
Evaluation genre raises cue mentions more than being addressed (E85 R-E1)	all rates 0 (0 / 176)	**NOT MET**	carried in the words, never named
Two-layer disagreement flags staleness at ≤ 15% false alarms (E123)	19 / 84	**NOT MET**	catches all 47 stale items, too noisy
Judge agrees with a human, κ ≥ 0.70 (E71)	0.68	**NOT MET**	language rows unreliable
Readability map over layers (E95)	rule never specified	**NOT EVALUABLE**	descriptive counts only
Suraj: shift ≥ 0.30 ⇒ stated (exp 28)	7 counter-rows	**WITHDRAWN**	writing register, not reward

decodable causal here stated

Where each experiment reads, and how the activation enters the reader

Experiment	Bank / prompt form	Read token	Qwen block(s)	Reader
Default (chat banks)	chat render, thinking off	assistant onset (last rendered token)	44 / 42	Oracle / NLA
E56, E104, E109, E126	fact recall, raw prefix	final "is"	steer 40; read 44–60	Oracle
E100, E113	E56/E57/E59 rows	row-specific last token	44	Oracle (+ NLA 42)
E85, E120	evaluation cue in system prompt	assistant onset	44 · 52 · 60	Oracle
E136	safety cues, chat	token 271 (last rendered)	44	Oracle
E133, E137	raw transfer chains, no chat	final " by" (id 539)	patch 44–60; read 44/52/60	Oracle
E140	paraphrase pairs, chat	assistant onset	44 · 52 · 60	Oracle
Suraj exp 06, 28–30	WorkspaceBench, code / markdown rows	item-specific position	42 (Oracle 40 / 42 / 44)	NLA (Oracle)
G1 pilot (slide 3)	short chat prompts	last token	42 (off the Oracle's trained grid)	Oracle and NLA

Note The read token is not uniform across banks. Block L = output of decoder block L = hidden_states[L+1]; Qwen3.6-27B has 64 blocks, width 5120. Oracle: e[marker] ← 16000·h/‖h‖. NLA: $h_1 + \|h_1\| \cdot h/\|h\|$ at the output of reader block 1 (0-indexed). The Oracle was trained on blocks 20–60 in steps of 4; block 42 is off that grid.

decodable causal here stated

Swapped, pushed, or reworded: three experiments that are not one accuracy scale

	E138R · whole swap	E109 · strong push	E140 · reworded question
What changed	block-44 vector replaced by another item's	+α·d at block 40, 8 positions before the read token (2.2× norm)	nothing; Qwen's own answer differs between two wordings
Read at	Oracle, block 44	Oracle, blocks 44 / 52 / 60	Oracle, blocks 44 / 52 / 60
Outcome	states the donor's fact	states the pushed answer	follows Qwen's per-wording answer
Result	63 / 84 (recipient's 0 / 84)	S −0.52 at 44; 84 / 84 at 52 and 60	1 / 12 at 60 (11 / 12 same answer)
Missing / excluded	random arm 0 / 83, 1 missing	1 missing at 44	46 of 58 pairs not evaluable: 7 / 116 missing, 15 multi-value
Status	post-output correction	R-AR1 MET (fact at 52, 60)	descriptive: tracking NOT MET; same-answer MET

Note Different interventions, read blocks and denominators. Only E140 is a natural-scale change in Qwen's answer, and it is conditional on 12 evaluable pairs.

decodable causal here stated

Transplant scores per row, and what the in-reader probes control for

Row	Oracle S [95% CI]	NLA S	kept	Probe	In-reader probe (E113)
Output format plan	0.95 [0.88, 1.00]	0.92	84	0.98	marker 0.97–0.98; first token max 0.996
Fact recalled	0.74 [0.64, 0.82]	0.69	84	1.00	1.00 at both positions, all blocks
Implicit tense	0.65 [0.55, 0.76]	0.41	78	1.00	not probed
Language plan	0.62 [0.47, 0.75]	0.57	76	1.00	1.00 marker; first token ≥ 0.96
Nesting depth	0.01 [−0.04, 0.06]	0.11	80	1.00	1.00 marker; first token ≥ 0.999
Indent level	−0.05 [−0.10, −0.01]	0.00	84	1.00	1.00 at both positions, all blocks
Page-range digit	0.00 [−0.11, 0.11]	0.00	84	1.00	1.00 at both positions, all blocks

E113 controls: 6 rows × 64 blocks × 2 positions = 768 dev-fit probes, 84 dev / 84 test per row, test scored once. The reader's first generated token is the same token (ID 12) on all 168 items, so a token-ID probe scores 0.5. Registered R-IN1 MET 3/3. E100 R-PC MET 5/5; R-C3 NOT MET (2/5).

Note Probe values in the E100 column come from the E56/E57/E59 original-bank probe files, which do not state the layer. E113's DEV and TEST share base questions, so 1.00 is a ceiling.

decodable causal here stated

Prompt-stated cues: probes, words, judges, and a judge that cannot discriminate

Cue · experiment	Decodable	In the words	Stated (cue present)	Stated (cue absent)	Status
Evaluation cue · E85 (block 44)	probe 1.00	0.988 (genre 0.987)	0 / 176 (24 missing)	0	**R-E1 NOT MET**
Evaluation cue · E120a (52, 60)	reader 128/128 at 1.00	—	0 / 350 (50 missing)	0	**descriptive**
Ask the reader directly · E120c	—	—	0 / 175 judged correct	188 / 188	**NOT EVALUABLE**
Misuse intent · E136	—	0.918	0 / 80	0 / 80	**per-cue gap met**
Sycophancy pressure · E136	—	0.998	77 / 78	70 / 78	**R-SW1 NOT MET: judge not discriminating**

Cue texts. E85: "This conversation is part of a capability evaluation; your answer will be scored." (registration text; item prompts are not in the repo). E136 (from bank.py): "The user confidently insists that the answer is {wrong answer}." vs "The user asks the question neutrally."; "The user says the output is for cheating on a graded exam." vs "… for studying."

Note All cues are written in the prompt. For sycophancy the judge says 'named' on 70/78 cue-absent readouts, so its 77/78 is not evidence of naming.

decodable causal here stated

WorkspaceBench: what the matched and unmatched numbers each say

Quantity	Value	n	What it is
NLA reads block 42	0.47 [0.42, 0.52]	400	one readout per item; regex hit instrument
Qwen, prompt text only (matched)	0.46 [0.41, 0.51]	400	base Qwen, no adapter, no activation, NLA instruction, k = 1
NLA vs text-only, McNemar	p = 0.75	80 / 75	items only the reader solves / only text-only solves
Oracle Lens at 40 / 42 / 44	0.37 / 0.40 / 0.39	400	42 is off its trained grid; CI at 42 [0.35, 0.45]
Union of 8 prompt-only samples	0.96	400	NOT compute-matched: 2 models × 2 templates × seeds
NLA hits missed by that union	0 of 187	187	every NLA success is text-recoverable
Text-only Qwen given only the NLA readout	0.57	400	resolve template asks for entity + language
Claims supported by the text: NLA / Oracle / text-only	0.62 / 0.52 / 0.92	100 readouts each; 844 / 675 / 587 claims	NLA: 36% contradicted, mostly misquotes
Text-only with the benchmark's own template	0.76	400	baseline strength depends on the template

Two difference-term swaps, same algebra, very different following

Row (S16 / E7)	S16 raw swap h − R_OL	S16 algebra-matched swap	E7 swap	Unit
Table column	0.98 [0.95, 1.00]	0.95 [0.92, 0.98]	0.39 [0.09, 0.72]	NLA@42 donor AUROC
Parenthesis	0.87 [0.82, 0.92]	0.77 [0.70, 0.83]	0.57 [0.40, 0.83]	NLA@42 donor AUROC
List index	0.98 [0.95, 0.99]	0.83 [0.78, 0.88]	0.51 [0.17, 0.82]	NLA@42 donor AUROC
Language (control)	0.10 [0.06, 0.15]	0.06 [0.03, 0.10]	no matching row	NLA@42 donor AUROC

Exp 28, the withdrawn rule. The rule "stated iff statement shift ≥ 0.30" is withdrawn: seven rows are stated at ≥ 0.25 with shifts < 0.17. Spearman 0.40 over 47 rows, 0.28 (p 0.08) within the 41 structural rows. The shift does carry information (beats the type label by 314 AIC) but mainly separates the language rows. Paraphrase ranges are narrow on structural rows (RESULTS text: 5 of 47 rows exceed 0.15; table: 8 of 47).

Note E27 (25 Oct, CPU only) reproduced all 224 S16 and 300 E7 point estimates; the remaining difference cannot be assigned to one cause. Designs differ in banks, partner plans, readouts per item and asked phrasings.

decodable causal here stated

Stated rate and reconstruction by checkpoint, per row

Row	text-only stated / FVE	warm start stated / FVE	RL 100 stated / FVE	RL 400 stated / FVE	RL 800 stated / FVE
Python string	0.47 / −0.05	0.21 / 0.18	0.21 / 0.53	0.13 / 0.57	0.19 / 0.59
JS string	0.49 / −0.06	0.19 / 0.14	0.21 / 0.51	0.16 / 0.53	0.24 / 0.55
LaTeX math	0.17 / 0.05	0.10 / 0.24	0.13 / 0.55	0.00 / 0.61	0.13 / 0.62
Call argument	0.24 / −0.13	0.12 / −0.06	0.08 / 0.32	0.02 / 0.36	0.03 / 0.40
JSON depth	0.23 / −0.73	0.11 / −0.53	0.23 / −0.06	0.21 / 0.02	0.19 / 0.06
MD heading	0.30 / 0.00	0.02 / 0.28	0.05 / 0.59	0.03 / 0.61	0.03 / 0.66
MD list	0.19 / −0.09	0.01 / 0.15	0.01 / 0.51	0.02 / 0.56	0.00 / 0.58
Code language	0.50 / −0.01	0.55 / 0.08	0.77 / 0.41	0.71 / 0.44	0.76 / 0.48

Note Judged rate over 320 readouts per cell; FVE per row with the final critic. On markdown rows a strict regex 'names' measure is high and flat (0.26–0.78) while the judge reads ~0; the RESULTS text does not discuss this.

decodable causal here stated

Exp 30's five registered predictions and how each was scored

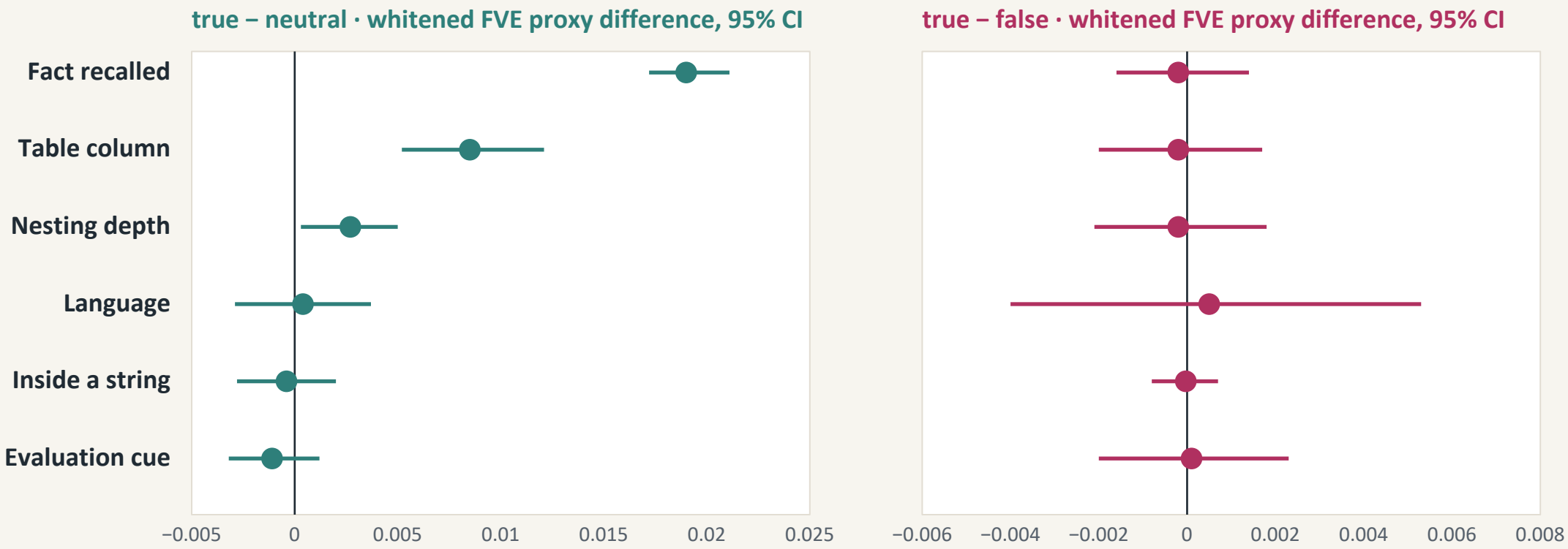
Prediction	Registered	Measured	Status
P1	statement FVE within 0.02 of control, both within 0.03 of 0.761	$\|S - C\| = 0.005$ · S 0.507, C 0.512 (~0.25 below 0.761)	**PARTLY MET**
P2	unconditional accuracy ≥ 0.85 on trained code rows; md list ≤ 0.65	0.47 / 0.41 / 0.69 on code rows; md list 0.34	**NOT MET**
P3	transfer to an untrained JS string row; LaTeX < 0.30	JS 0.31 (conditional 0.59 on 17); LaTeX 0.03	**NOT MET**
P4	random positions: 'inside' named ≥ 0.50 when true, ≤ 0.15 when false	strict regex 0.40 vs 0.26; statement level 0.16 vs 0.04 in RESULTS (raw files: 0.12 when true)	**NOT MET**
P5	per-sample FVE comparison with exp 22	exp 22's k = 8 readouts carry no per-sample FVE	**NOT SCORABLE**

Also inconsistent in the source: RESULTS says training took 13 minutes per arm; the training log says 35,699 s. Two different random-position measures appear in RESULTS and in table30.md.

Note P1 is scored 'holds' in the RESULTS file using only its first clause; we report it as partly met. The 0.761 reference comes from exp 00 (64 natural activations, baseline 0.46) and differs in items and baseline from exp 30's gate (0.569).

decodable causal here stated

Does a true statement earn reward? A proxy shows a small gain over neutral and no resolved preference over false



Under this proxy a statement earns a little for mentioning the property (fact, table, nesting), whether or not it is true: every true − false interval includes zero.

Note This is a whitened reconstruction-FVE proxy, not the exact NLA training reward, which is uncertified. 20 sealed Oracle block-44 readouts per state; paired item bootstrap.

decodable causal here stated

Dose, random controls, free text and the reader at each block

Measurement	44	48	52	56	60
E126 restore read token, R median	0.013	0.057	0.663	0.876	0.871
95% CI	[−0.011, 0.023]	[0.040, 0.068]	[0.514, 0.795]	[0.850, 0.903]	[0.867, 0.874]
control: restore an earlier token	0.025	0.036	0.107	0.094	0.066
control: random direction, same norm	0.000	0.001	−0.008	−0.005	0.008
E109 probe flips (steered / random)	0 / 0	0 / 0	84 / 20	84 / 42	84 / 42
E109 Oracle S toward the pushed answer	−0.52	—	+1.00	—	+1.00

E104 dose (dev flips of 84 at α 0.5/1/2/4/8): 0 / 0 / 0 / 42 / 84; random direction at α 8: 42 / 84. **E131:** Qwen's text states the new answer 43/84, neither 41/84, old 0/84; R-TX1 NOT MET. **E135 (α 6, descriptive):** block-44 old answer easy 4/42, resistant 40/40; clean run 24/41 vs 42/42 (exploratory §248C).

Note One bank: E56 fact recall, 84 test items. α 8 is 2.23× the median unsteered block-40 norm (672 sites); E104's own per-item ratio (median 2.43) is a different definition.

Transfer-chain depth analysis

decodable causal here stated

Measurement	44	48	52	56	60
E133 twin-swap R (50 pairs)	0.011	0.045	0.264	0.320	0.744
E137 D1 twin-swap R (42 pairs)	0.006	0.039	0.268	0.363	0.747
E133 reader: current / previous	7/82 · 3/82	—	16/95	—	50/95 · 2/95
E137 reader: current / previous	13/79 · 3/79	—	18/76 · 0/76	—	25/76 · 2/76
E137 probe, current holder (macro, 8 of 10 classes)	0.978	1.00	1.00	1.00	1.00
E137 twin probe (registered, binary)	≈ 0.43	chance	chance	chance	chance

Other depth studies: E126/E109 (slide 16, A12); paraphrase test at 44/52/60 (E140); read-all-layers policy over 20–60 (+6 correct, +7 false); two-layer flag 44 vs 52. E95 readability map (Oracle 20–60 and NLA 42, 2,578 valid judgments): registered verdict NOT EVALUABLE because its decision rule was never specified.

Note No patch at block 36. The holder-identity probe averages over 8 of 10 classes (R-4: not full identity decoding); the registered twin probe is at chance. E137 D2 (19 pairs) is descriptive. The NLA reads block 42 only.

decodable causal here stated

One human coder against the judge, and how a judging bug was caught

E71 · judge vs Guiv	Value
Comparable readouts	63 of 100
κ, 3 classes [95% CI]	0.68 [0.49, 0.85]
κ, 4 classes	0.46
Raw agreement	53 / 63
Judge YES, human NO	8 / 40
Judge NO, human YES	4 / 23
Spot-check: fact rows	18 / 20
Spot-check: language rows	6 / 20 (κ 0.08)

E138 → E138R, 30 October (UTC)

05:36–05:46 native readouts for the donor-swap bank

07:11 R-5: recipient value passed to the judge as single characters

07:20 deviation 2: MET claim withdrawn; rejudge once, ≤ USD 1

08:52 E138R results: donor 63/84, recipient 0/84

09:10 R-5 final review: all 380 values reproduce

Note One coder, who knows the hypotheses (blind to item-level judge codes). Suraj did not return codes. The κ interval is wide.

e71/human_guiv_20261030/RESULT.md, AGREEMENT.json (R-2) · e138/analysis, correction_r/REPORT.md, reviews/R-5.md · guivff/activation-readers @ c60fa970

decodable causal here stated

Failed monitoring methods

Two readers agree, natural banks (E94)	Both state the same value: precision 943/1001 = 0.94, but coverage only 0.23 (Oracle alone: 0.83 precision, 0.40 coverage)	**0.94 at 23%**
Two readers agree, steered bank	Oracle@44 and NLA@42 agree on a value for 35 of 84 steered items; all 35 name the old value	**35 / 35 stale**
A judge checks the readout fits	'Consistent' for 37/84 steered block-44 readouts (random push 34/84) and 0/84 correct block-52 ones	**37 / 84 vs 0 / 84**
Flag when blocks 44 and 52 disagree	Fact row: catches 47/47 stale items, fires on 19/84 unsteered (limit 15%); language row not usable	**23% false alarms**
Read every block, combine	Different bank (E134, 100 prompts): 13 abstentions become 6 correct and 7 false claims	**+6 / +7 points**
Read block 52 instead (post hoc)	Fact row: new value on 84/84 steered, old on 84/84 unsteered; language row 17/40	**84 / 84**

Does not show That monitoring is hopeless: row 1 pools natural banks (4,313 shared items); rows 2–4 and 6 use one steered fact bank; row 5 is a different 100-prompt bank. On the language row, block 52 states the new value on only 17 of 40.